

Modern Practical Control Systems

Section 0

Mathematical Foundations

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Chapter 1

Complex Numbers and Imaginary Arithmetic

Complex numbers are fundamental to control systems analysis. They appear naturally in the solution of differential equations, in the Laplace transform, in frequency response analysis, and in stability analysis.

1.1 Definition and Representation

Definition 0.1: Complex Number

A **complex number** z is defined as:

$$z = a + jb \quad (1.1)$$

where $a, b \in \mathbb{R}$, and $j = \sqrt{-1}$ is the **imaginary unit**. The real part is $\text{Re}(z) = a$ and the imaginary part is $\text{Im}(z) = b$.

Note 1.1. In electrical engineering, j is often used instead of i for the imaginary unit to avoid confusion with current. We adopt this convention throughout.

Complex numbers can be represented in multiple forms:

Rectangular Form

$$z = a + jb \quad (1.2)$$

Polar Form

$$z = r(\cos \theta + j \sin \theta) = r \angle \theta \quad (1.3)$$

where $r = |z| = \sqrt{a^2 + b^2}$ is the **magnitude** (or modulus) and $\theta = \arg(z) = \arctan(b/a)$ is the **argument** (or phase angle).

Exponential Form (Euler's Formula)

$$z = re^{j\theta} \quad (1.4)$$

Euler's formula, one of the most elegant results in mathematics, states:

$$\boxed{e^{j\theta} = \cos \theta + j \sin \theta} \quad (1.5)$$

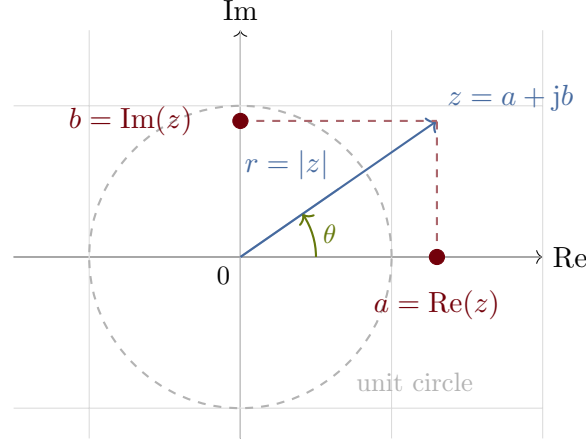


Figure 1.1: Geometric representation of a complex number in the complex plane (Argand diagram). The blue vector shows the complex number z , with dashed red projections onto the real and imaginary axes. The green angle θ represents the argument, and the unit circle (dashed gray) provides a reference. Colors: Atlantic (blue) for magnitude, Garnet (red) for projections, Horseshoe (green) for angle.

1.2 Arithmetic Operations

Addition and Subtraction

$$(a + jb) \pm (c + jd) = (a \pm c) + j(b \pm d) \quad (1.6)$$

Multiplication

$$(a + jb)(c + jd) = (ac - bd) + j(ad + bc) \quad (1.7)$$

In polar form: $r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$

Division

$$\frac{a + jb}{c + jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)} = \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2} \quad (1.8)$$

In polar form: $\frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$

Complex Conjugate

$$z^* = \bar{z} = a - jb \quad (1.9)$$

Important property: $z \cdot z^* = |z|^2 = a^2 + b^2$

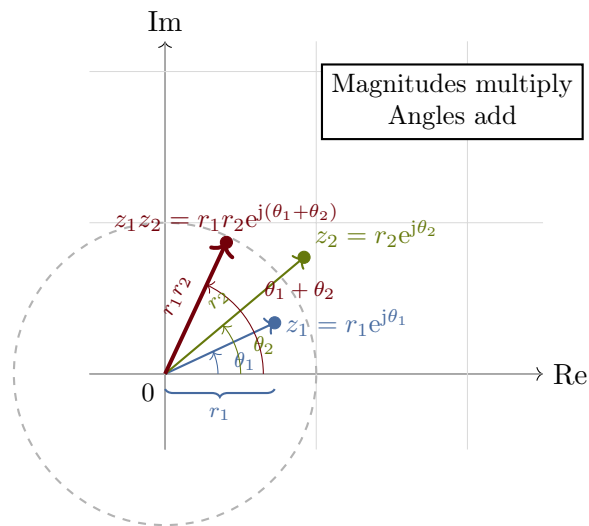


Figure 1.2: Complex multiplication in polar form. When multiplying two complex numbers $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$, the magnitudes multiply ($r_1 r_2$) and the angles add ($\theta_1 + \theta_2$). This geometric interpretation makes polar form ideal for multiplication and division operations.

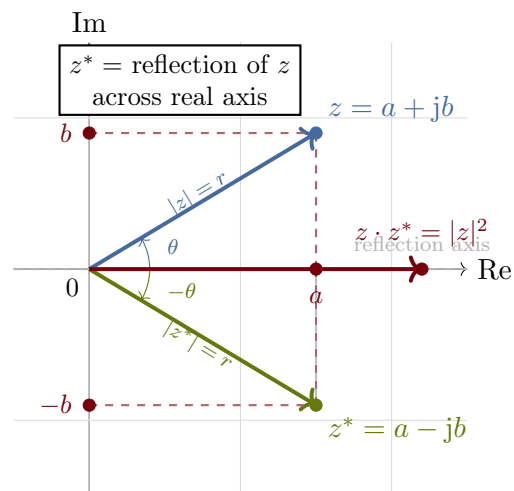


Figure 1.3: Complex conjugate visualization. The conjugate $z^* = a - jb$ is the reflection of $z = a + jb$ across the real axis. Both have the same magnitude $|z| = |z^*| = r$, but opposite angles $\pm\theta$. The product $z \cdot z^* = |z|^2 = a^2 + b^2$ is always a real, non-negative number lying on the real axis.

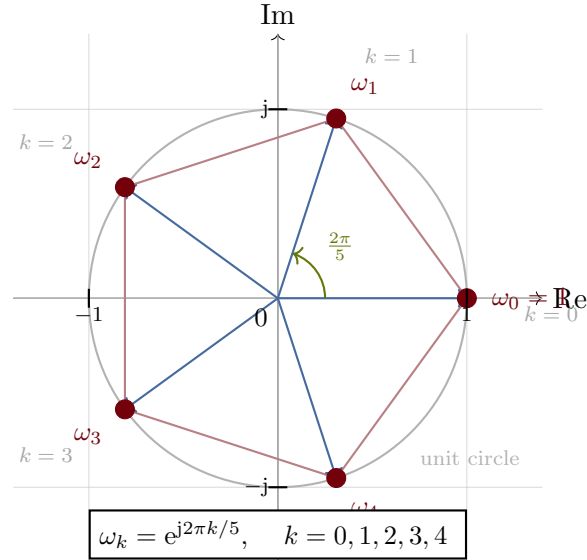


Figure 1.4: The fifth roots of unity. The n th roots of unity are n complex numbers evenly distributed on the unit circle, satisfying $z^n = 1$. For $n = 5$, these roots $\omega_k = e^{j2\pi k/5}$ are separated by angles of $2\pi/5 = 72$ and form a regular pentagon. The root $\omega_0 = 1$ always lies on the positive real axis.

Powers (De Moivre's Theorem)

$$z^n = r^n e^{jn\theta} = r^n (\cos n\theta + j \sin n\theta) \quad (1.10)$$

Roots

The n th roots of $z = re^{j\theta}$ are:

$$z^{1/n} = r^{1/n} e^{j(\theta+2\pi k)/n}, \quad k = 0, 1, \dots, n-1 \quad (1.11)$$

Example 0.1: Complex Number Operations

Find the polar form, magnitude, and phase of $z = 3 + 4j$.

Solution:

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \quad (1.12)$$

$$\theta = \arctan\left(\frac{4}{3}\right) \approx 53.13 = 0.927 \text{ rad} \quad (1.13)$$

$$z = 5e^{j(0.927)} = 5\angle 53.13 \quad (1.14)$$

Example 0.2: Multiplication and Division in Polar Form

Given $z_1 = 2e^{j\pi/6}$ and $z_2 = 3e^{j\pi/4}$, compute $z_1 \cdot z_2$ and z_1/z_2 .

Solution:

For multiplication, we multiply magnitudes and add angles:

$$z_1 \cdot z_2 = (2)(3)e^{j(\pi/6+\pi/4)} = 6e^{j(5\pi/12)} \quad (1.15)$$

Converting to rectangular form: $z_1 \cdot z_2 = 6 \cos(5\pi/12) + 6j \sin(5\pi/12) \approx 1.55 + 5.80j$.

For division, we divide magnitudes and subtract angles:

$$\frac{z_1}{z_2} = \frac{2}{3} e^{j(\pi/6 - \pi/4)} = \frac{2}{3} e^{-j\pi/12} \quad (1.16)$$

Converting to rectangular form: $z_1/z_2 = \frac{2}{3} \cos(-\pi/12) + \frac{2}{3}j \sin(-\pi/12) \approx 0.644 - 0.173j$.

Key insight: Polar form makes multiplication and division much simpler than rectangular form, which is why it is preferred in frequency response analysis.

Example 0.3: Finding All Cube Roots of a Complex Number

Find all cube roots of $z = 8$.

Solution:

First, write $z = 8$ in polar form: $z = 8e^{j \cdot 0} = 8e^{j \cdot 2\pi k}$ for any integer k .

The n th roots of $z = re^{j\theta}$ are given by:

$$z^{1/n} = r^{1/n} e^{j(\theta + 2\pi k)/n}, \quad k = 0, 1, \dots, n-1$$

For the cube roots ($n = 3$) of $8e^{j \cdot 0}$:

$$z_0 = 8^{1/3} e^{j(0 + 2\pi \cdot 0)/3} = 2e^{j \cdot 0} = 2 \quad (1.17)$$

$$z_1 = 8^{1/3} e^{j(0 + 2\pi \cdot 1)/3} = 2e^{j2\pi/3} = 2 \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = -1 + j\sqrt{3} \quad (1.18)$$

$$z_2 = 8^{1/3} e^{j(0 + 2\pi \cdot 2)/3} = 2e^{j4\pi/3} = 2 \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) = -1 - j\sqrt{3} \quad (1.19)$$

Verification: We can verify $(2)^3 = 8$, $(-1 + j\sqrt{3})^3 = 8$, and $(-1 - j\sqrt{3})^3 = 8$.

Geometric interpretation: The three roots are equally spaced on a circle of radius 2 in the complex plane, separated by 120 (i.e., $2\pi/3$ radians).

Example 0.4: Deriving Trigonometric Identities Using Euler's Formula

Use Euler's formula to derive the double-angle formulas for $\cos(2\theta)$ and $\sin(2\theta)$.

Solution:

From Euler's formula: $e^{j\theta} = \cos \theta + j \sin \theta$.

Consider $e^{j(2\theta)}$. We can write this in two ways:

Method 1: Direct application of Euler's formula:

$$e^{j(2\theta)} = \cos(2\theta) + j \sin(2\theta)$$

Method 2: Using properties of exponents:

$$\begin{aligned} e^{j(2\theta)} &= \left(e^{j\theta} \right)^2 = (\cos \theta + j \sin \theta)^2 \\ &= \cos^2 \theta + 2j \cos \theta \sin \theta + j^2 \sin^2 \theta \\ &= (\cos^2 \theta - \sin^2 \theta) + j(2 \cos \theta \sin \theta) \end{aligned}$$

Equating real and imaginary parts from both methods:

$$\text{Re : } \cos(2\theta) = \cos^2 \theta - \sin^2 \theta \quad (1.20)$$

$$\text{Im : } \sin(2\theta) = 2 \cos \theta \sin \theta \quad (1.21)$$

Extension: This technique can be used to derive any multiple-angle formula. For example, expanding $(e^{j\theta})^3$ yields the triple-angle formulas.

1.3 Properties of Complex Operations

This section provides formal proofs of fundamental properties of complex number operations. These properties are essential for understanding frequency response analysis and stability theory.

Key Property: Magnitude of a Product

For any two complex numbers z_1 and z_2 :

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2| \quad (1.22)$$

The magnitude of a product equals the product of the magnitudes.

Proof. Let $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$ be two complex numbers in polar form, where $r_1 = |z_1|$ and $r_2 = |z_2|$.

The product is:

$$z_1 \cdot z_2 = r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

Taking the magnitude:

$$|z_1 \cdot z_2| = |r_1 r_2 e^{j(\theta_1 + \theta_2)}| = r_1 r_2 = |z_1| \cdot |z_2|$$

where we used the fact that $|e^{j\phi}| = 1$ for any real ϕ . □

Key Property: Argument of a Product

For any two complex numbers z_1 and z_2 (with $z_1, z_2 \neq 0$):

$$\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2) \pmod{2\pi} \quad (1.23)$$

The argument of a product equals the sum of the arguments (modulo 2π).

Proof. Let $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$, where $\theta_1 = \arg(z_1)$ and $\theta_2 = \arg(z_2)$.

The product is:

$$z_1 \cdot z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

By definition of the argument (the angle in polar form):

$$\arg(z_1 \cdot z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2) \pmod{2\pi}$$

The modulo 2π accounts for the fact that angles differing by 2π represent the same direction. □

Key Property: Magnitude Squared via Conjugate

For any complex number z :

$$|z|^2 = z \cdot z^* \quad (1.24)$$

The squared magnitude equals the product of a number and its complex conjugate.

Proof. Let $z = a + jb$ where $a, b \in \mathbb{R}$.

The complex conjugate is $z^* = a - jb$.

Computing the product:

$$\begin{aligned} z \cdot z^* &= (a + jb)(a - jb) \\ &= a^2 - jab + jab - j^2 b^2 \\ &= a^2 - (-1)b^2 \\ &= a^2 + b^2 \end{aligned}$$

By definition of magnitude: $|z|^2 = (\sqrt{a^2 + b^2})^2 = a^2 + b^2$.

Therefore: $|z|^2 = z \cdot z^*$.

Alternative proof using polar form: Let $z = re^{j\theta}$. Then $z^* = re^{-j\theta}$, and:

$$z \cdot z^* = re^{j\theta} \cdot re^{-j\theta} = r^2 e^{j \cdot 0} = r^2 = |z|^2$$

□

Note 1.2. The property $|z|^2 = z \cdot z^*$ is particularly useful in control systems when computing power spectral densities and when rationalizing complex-valued transfer functions for frequency response analysis.

1.4 Complex Numbers in Control Systems

Complex numbers appear throughout control theory:

- **Poles and Zeros:** The roots of transfer function polynomials are generally complex.
- **Eigenvalues:** System matrices often have complex eigenvalues that determine stability and oscillatory behavior.
- **Frequency Response:** Evaluating transfer functions at $s = j\omega$ yields complex numbers representing magnitude and phase.
- **Sinusoidal Steady-State:** Complex exponentials simplify analysis of sinusoidal inputs.

Chapter 2

Calculus Review

2.1 Differential Calculus

2.1.1 Derivatives and Rules

The derivative of a function $f(t)$ represents its instantaneous rate of change:

$$\frac{df}{dt} = f'(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad (2.1)$$

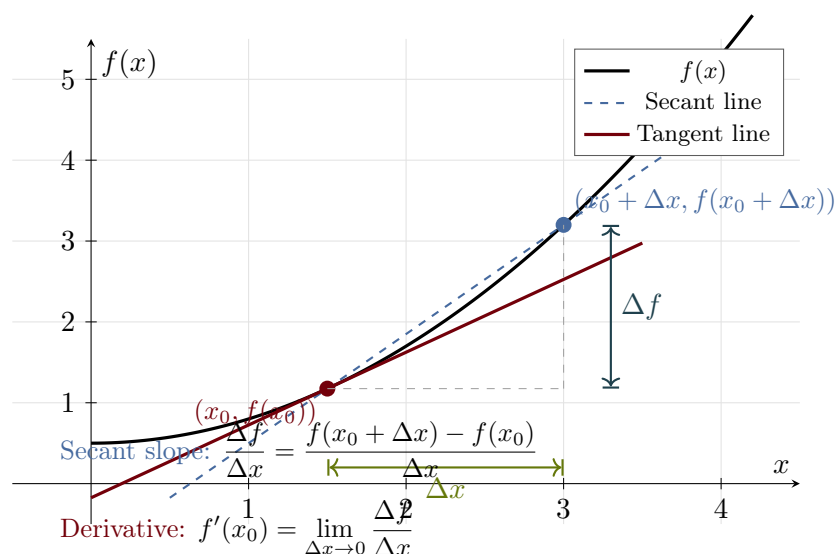


Figure 2.1: Visualization of the derivative as the limit of secant line slopes. The curve $f(x)$ is shown in black. The secant line (dashed Atlantic blue) connects two points separated by Δx , with slope $\Delta f/\Delta x$. As $\Delta x \rightarrow 0$, the secant line approaches the tangent line (Garnet red) at x_0 , whose slope equals the derivative $f'(x_0)$. The intervals Δx (Horseshoe green) and Δf (Congaree teal) are marked.

Rule	Formula
Power Rule	$\frac{d}{dt}(t^n) = nt^{n-1}$
Exponential	$\frac{d}{dt}(e^{at}) = ae^{at}$
Logarithm	$\frac{d}{dt}(\ln t) = \frac{1}{t}$
Sine	$\frac{d}{dt}(\sin \omega t) = \omega \cos \omega t$
Cosine	$\frac{d}{dt}(\cos \omega t) = -\omega \sin \omega t$
Product Rule	$\frac{d}{dt}(fg) = f'g + fg'$
Quotient Rule	$\frac{d}{dt}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$
Chain Rule	$\frac{d}{dt}f(g(t)) = f'(g(t)) \cdot g'(t)$

Table 2.1: Essential differentiation rules for control systems.

Key Differentiation Rules

Product Rule Application

Differentiate $f(t) = t^2 e^{-at}$ where $a > 0$ is a constant.

Solution: Using the product rule with $u = t^2$ and $v = e^{-at}$:

$$\frac{df}{dt} = \frac{d}{dt}(t^2) \cdot e^{-at} + t^2 \cdot \frac{d}{dt}(e^{-at}) \quad (2.2)$$

$$= 2t \cdot e^{-at} + t^2 \cdot (-ae^{-at}) \quad (2.3)$$

$$= e^{-at}(2t - at^2) \quad (2.4)$$

$$= te^{-at}(2 - at) \quad (2.5)$$

This type of function appears frequently in control systems as a damped response with a time-varying amplitude.

2.1.2 Partial Derivatives

For functions of multiple variables $f(x_1, x_2, \dots, x_n)$, partial derivatives hold all variables constant except one:

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, \dots, x_i + \Delta x_i, \dots, x_n) - f(x_1, \dots, x_n)}{\Delta x_i} \quad (2.6)$$

The **gradient** is the vector of all partial derivatives:

$$\nabla f = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \dots \quad \frac{\partial f}{\partial x_n} \right]^\top \quad (2.7)$$

The **Hessian** is the matrix of second partial derivatives:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (2.8)$$

2.1.3 Taylor Series Expansion

Taylor series are essential for linearization of nonlinear systems:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots \quad (2.9)$$

For multivariate functions about point $\mathbf{x}_0$:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) \quad (2.10)$$

Linearization in Control

For nonlinear systems $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, we linearize about an equilibrium point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$\Delta \dot{\mathbf{x}} \approx \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \Delta \mathbf{u} \quad (2.11)$$

where $\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)}$ and $\mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)}$

Taylor Expansion of Sine Function

Find the Taylor expansion of $\sin(x)$ about $x = 0$ (Maclaurin series) to fourth order.

Solution: We need the derivatives of $f(x) = \sin(x)$ evaluated at $x = 0$:

$$f(x) = \sin(x) \quad f(0) = 0 \quad (2.12)$$

$$f'(x) = \cos(x) \quad f'(0) = 1 \quad (2.13)$$

$$f''(x) = -\sin(x) \quad f''(0) = 0 \quad (2.14)$$

$$f'''(x) = -\cos(x) \quad f'''(0) = -1 \quad (2.15)$$

$$f^{(4)}(x) = \sin(x) \quad f^{(4)}(0) = 0 \quad (2.16)$$

Substituting into the Taylor series formula:

$$\sin(x) = 0 + 1 \cdot x + \frac{0}{2!}x^2 + \frac{-1}{3!}x^3 + \frac{0}{4!}x^4 + \dots = x - \frac{x^3}{6} + \mathcal{O}(x^5) \quad (2.17)$$

For small angles ($|x| \ll 1$), we often use the approximation $\sin(x) \approx x$, which is the first-order Taylor expansion. This is the “small angle approximation” commonly used in linearizing pendulum and rotational dynamics.

2.2 Integral Calculus

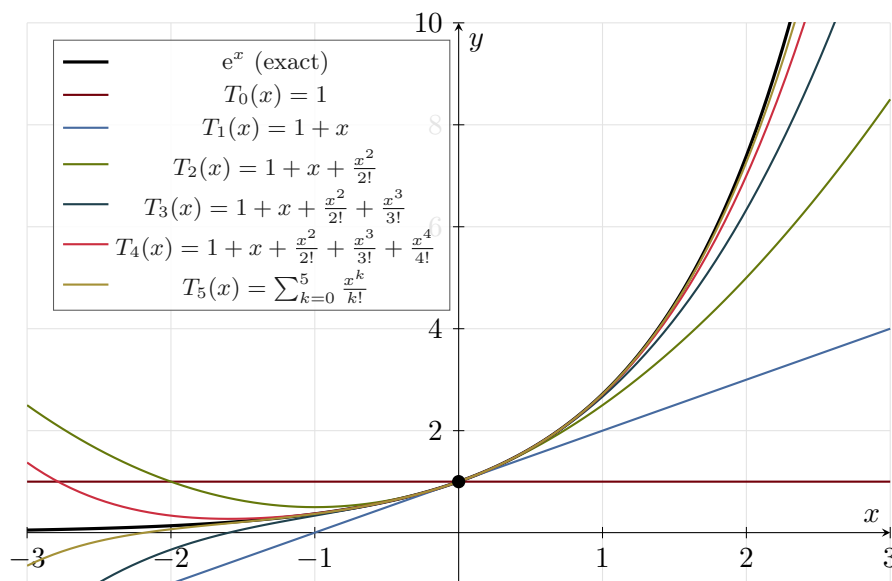
2.2.1 Definite and Indefinite Integrals

The indefinite integral (antiderivative):

$$\int f(t) dt = F(t) + C \quad \text{where} \quad F'(t) = f(t) \quad (2.18)$$

The definite integral:

$$\int_a^b f(t) dt = F(b) - F(a) \quad (2.19)$$



Taylor Series:
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Figure 2.2: Convergence of Taylor series approximations to e^x centered at $x = 0$. The exact function e^x is shown in black. Successive Taylor polynomial approximations $T_n(x) = \sum_{k=0}^n x^k/k!$ are shown for orders $n = 0$ (Garnet), $n = 1$ (Atlantic), $n = 2$ (Horseshoe), $n = 3$ (Congaree), $n = 4$ (Rose), and $n = 5$ (Honeycomb). Higher-order approximations provide better accuracy over a wider range of x values. All approximations are exact at the expansion point $x = 0$.

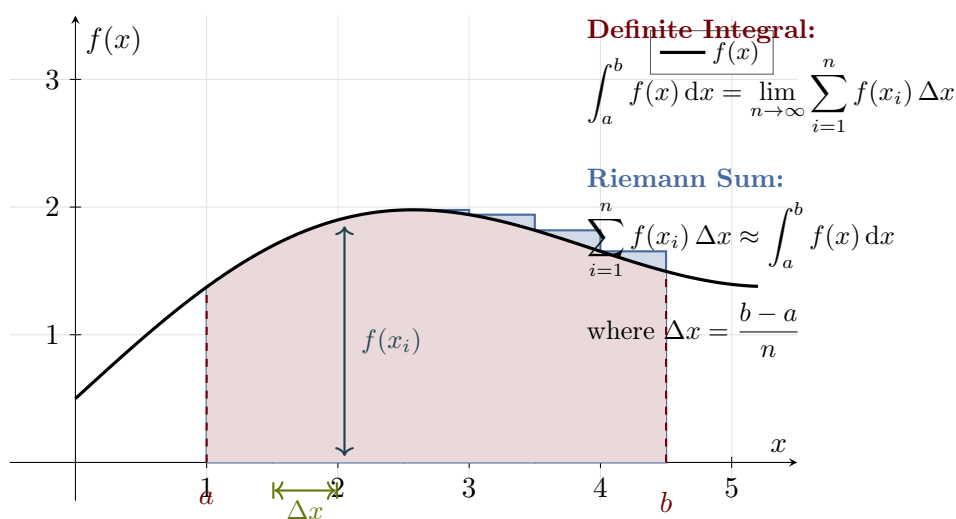


Figure 2.3: Visualization of the definite integral as area under a curve. The function $f(x)$ is shown in black. The shaded Garnet region represents the exact area $\int_a^b f(x) dx$. The Atlantic blue rectangles show a Riemann sum approximation with $n = 7$ rectangles of width $\Delta x = (b - a)/n$. As $n \rightarrow \infty$, the Riemann sum converges to the definite integral.

Key Integration Formulas

Function	Integral
t^n ($n \neq -1$)	$\frac{t^{n+1}}{n+1} + C$
$\frac{1}{t}$	$\ln t + C$
e^{at}	$\frac{1}{a}e^{at} + C$
$\sin \omega t$	$-\frac{1}{\omega} \cos \omega t + C$
$\cos \omega t$	$\frac{1}{\omega} \sin \omega t + C$
$\frac{1}{a^2+t^2}$	$\frac{1}{a} \arctan \frac{t}{a} + C$

Table 2.2: Essential integration formulas.

2.2.2 Integration by Parts

$$\int u \, dv = uv - \int v \, du \quad (2.20)$$

This technique is essential for deriving Laplace transform properties.

Integration by Parts

Evaluate the integral $\int te^{-t} \, dt$.

Solution: Choose $u = t$ and $dv = e^{-t} \, dt$. Then:

$$u = t \quad du = dt \quad (2.21)$$

$$dv = e^{-t} \, dt \quad v = -e^{-t} \quad (2.22)$$

Applying integration by parts:

$$\int te^{-t} \, dt = uv - \int v \, du \quad (2.23)$$

$$= t(-e^{-t}) - \int (-e^{-t}) \, dt \quad (2.24)$$

$$= -te^{-t} + \int e^{-t} \, dt \quad (2.25)$$

$$= -te^{-t} - e^{-t} + C \quad (2.26)$$

$$= -e^{-t}(t+1) + C \quad (2.27)$$

This integral appears when computing Laplace transforms of polynomial-exponential products.

2.2.3 Integration Techniques

Substitution (u-Substitution)

When an integrand contains a composite function, substitution can simplify the integral:

$$\int f(g(t))g'(t) dt = \int f(u) du \quad \text{where } u = g(t) \quad (2.28)$$

Choosing a Substitution

Look for an “inner function” $u = g(t)$ whose derivative $g'(t)$ also appears in the integrand. Common patterns include:

- $\int e^{at} \cdot a dt$: let $u = at$
- $\int \frac{f'(t)}{f(t)} dt$: let $u = f(t)$, giving $\ln |f(t)| + C$
- $\int t^n f(t^{n+1}) dt$: let $u = t^{n+1}$

Partial Fraction Decomposition

For rational functions $\frac{P(s)}{Q(s)}$ where $\deg(P) < \deg(Q)$, partial fractions decompose the integrand into simpler terms. This is essential for inverse Laplace transforms.

Case 1: Distinct real roots. If $Q(s) = (s - p_1)(s - p_2) \cdots (s - p_n)$:

$$\frac{P(s)}{Q(s)} = \frac{A_1}{s - p_1} + \frac{A_2}{s - p_2} + \cdots + \frac{A_n}{s - p_n} \quad (2.29)$$

Case 2: Repeated real roots. If $Q(s) = (s - p)^n$:

$$\frac{P(s)}{(s - p)^n} = \frac{A_1}{s - p} + \frac{A_2}{(s - p)^2} + \cdots + \frac{A_n}{(s - p)^n} \quad (2.30)$$

Case 3: Complex conjugate roots. If $Q(s)$ contains $(s^2 + as + b)$ with $a^2 - 4b < 0$:

$$\frac{P(s)}{s^2 + as + b} = \frac{As + B}{s^2 + as + b} \quad (2.31)$$

Cover-Up Method for Residues

For distinct real poles, the residue A_k can be found by:

$$A_k = \lim_{s \rightarrow p_k} (s - p_k) \frac{P(s)}{Q(s)} = \frac{P(p_k)}{Q'(p_k)} \quad (2.32)$$

This “cover-up” method is faster than solving systems of equations.

2.2.4 Convolution Integral

The convolution of two functions $f(t)$ and $g(t)$ is:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \quad (2.33)$$

For causal systems (signals zero for $t < 0$):

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \quad (2.34)$$

Convolution in Control Systems

The output $y(t)$ of a linear time-invariant (LTI) system with impulse response $h(t)$ to input $u(t)$ is:

$$y(t) = (h * u)(t) = \int_0^t h(\tau)u(t - \tau) d\tau \quad (2.35)$$

Convolution of Two Exponentials

Compute the convolution $(f * g)(t)$ for $t \geq 0$ where $f(t) = e^{-at}$ and $g(t) = e^{-bt}$ with $a \neq b$ and both signals are zero for $t < 0$.

Solution: Using the causal convolution formula:

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \quad (2.36)$$

$$= \int_0^t e^{-a\tau} e^{-b(t-\tau)} d\tau \quad (2.37)$$

$$= e^{-bt} \int_0^t e^{-a\tau} e^{b\tau} d\tau \quad (2.38)$$

$$= e^{-bt} \int_0^t e^{(b-a)\tau} d\tau \quad (2.39)$$

Evaluating the integral:

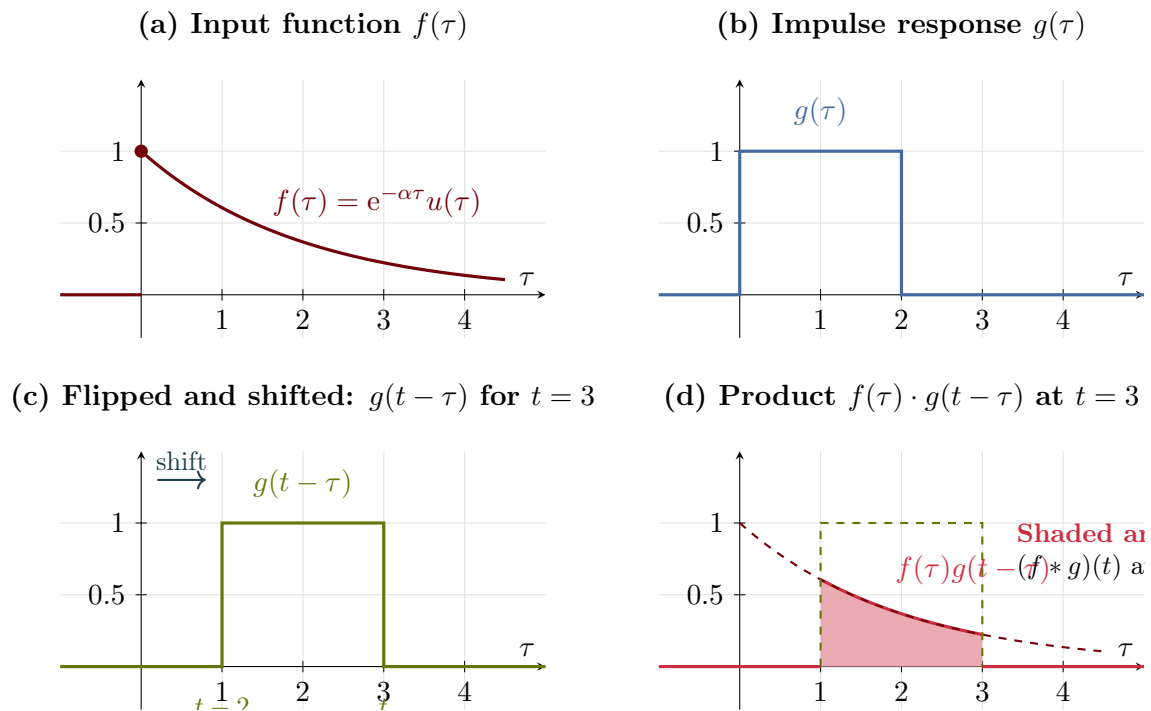
$$(f * g)(t) = e^{-bt} \left[\frac{e^{(b-a)\tau}}{b-a} \right]_0^t \quad (2.40)$$

$$= e^{-bt} \cdot \frac{e^{(b-a)t} - 1}{b-a} \quad (2.41)$$

$$= \frac{e^{-bt} e^{(b-a)t} - e^{-bt}}{b-a} \quad (2.42)$$

$$= \frac{e^{-at} - e^{-bt}}{b-a} \quad (2.43)$$

This result is symmetric in the sense that swapping a and b gives the same answer (note the sign change in both numerator and denominator). This convolution appears when analyzing cascaded first-order systems.



$$\text{Convolution Integral: } (f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

Figure 2.4: Visualization of the convolution operation $(f * g)(t)$. (a) The input function $f(\tau) = e^{-\alpha\tau}u(\tau)$ in Garnet. (b) The impulse response $g(\tau)$, a rectangular pulse, in Atlantic blue. (c) The flipped and shifted function $g(t - \tau)$ for $t = 3$ in Horseshoe green—convolution involves sliding this function across f . (d) The product $f(\tau) \cdot g(t - \tau)$ in Rose; the shaded area equals the convolution value $(f * g)(3)$. As t varies, this area traces out the complete convolution.

Chapter 3

Differential Equations

Differential equations describe how systems evolve over time and are the mathematical language of dynamics. In control systems, we encounter differential equations when modeling physical systems, analyzing stability, and designing controllers.

Definition: Ordinary Differential Equation (ODE)

An **ordinary differential equation** is an equation involving a function $y(t)$ and its derivatives with respect to a single independent variable t :

$$F\left(t, y, \frac{dy}{dt}, \frac{d^2y}{dt^2}, \dots, \frac{d^ny}{dt^n}\right) = 0 \quad (3.1)$$

The **order** of the ODE is the highest derivative that appears. An ODE is **linear** if the function y and all its derivatives appear linearly (to the first power and not multiplied together).

3.1 Ordinary Differential Equations (ODEs)

3.1.1 First-Order Linear ODEs

The general first-order linear ODE:

$$\frac{dy}{dt} + p(t)y = q(t) \quad (3.2)$$

Solution using Integrating Factor

Multiply by $\mu(t) = e^{\int p(t) dt}$:

$$y(t) = \frac{1}{\mu(t)} \left[\int \mu(t)q(t) dt + C \right] \quad (3.3)$$

For constant coefficients $\frac{dy}{dt} + ay = b$:

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a} \right) e^{-at} \quad (3.4)$$

Time Constant

For a first-order system $\tau \frac{dy}{dt} + y = Ku$, the **time constant** τ determines:

- The speed of response: smaller τ means faster response
- At $t = \tau$, the response reaches 63.2% of its final value
- At $t = 4\tau$, the response is within 2% of steady state
- The bandwidth is $\omega_{bw} = 1/\tau$ rad/s

Example: RC Circuit

An RC circuit with voltage source V_s satisfies:

$$RC \frac{dv_c}{dt} + v_c = V_s \quad (3.5)$$

With $v_c(0) = 0$, the capacitor voltage is:

$$v_c(t) = V_s (1 - e^{-t/RC}) \quad (3.6)$$

The time constant $\tau = RC$ determines how quickly the system responds.

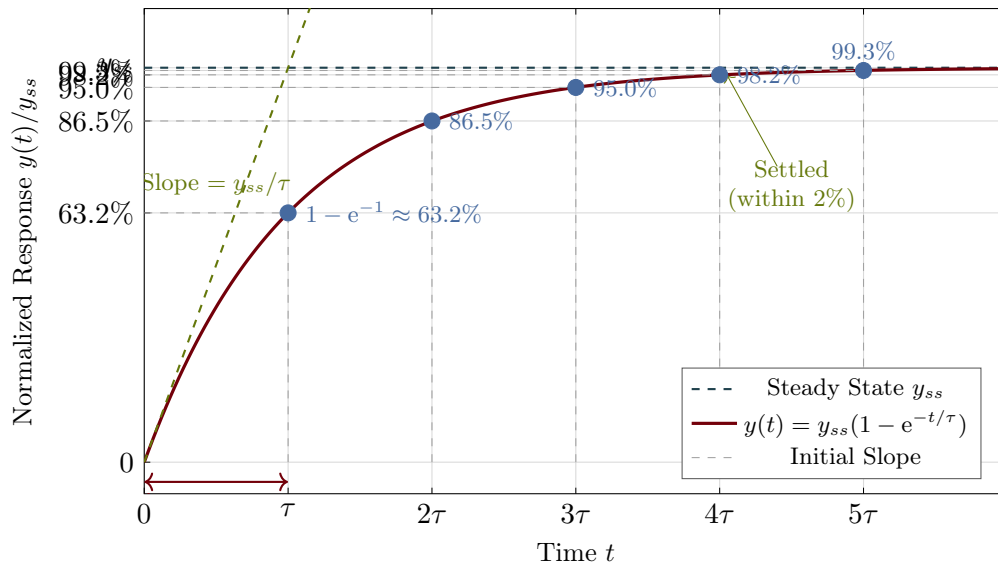


Figure 3.1: Step response of a first-order system $y(t) = y_{ss}(1 - e^{-t/\tau})$ showing the characteristic exponential approach to steady state. The time constant τ uniquely determines the response speed: at $t = \tau$, the response reaches 63.2% of the final value. Key milestones are marked at each multiple of τ : 63.2% at τ , 86.5% at 2τ , 95.0% at 3τ , 98.2% at 4τ , and 99.3% at 5τ . The initial tangent line (Horseshoe green) has slope y_{ss}/τ and would reach y_{ss} at $t = \tau$ if the rate were maintained. The system is typically considered settled (within 2% of y_{ss}) after 4τ .

3.1.2 Second-Order Linear ODEs with Constant Coefficients

The standard form:

$$\frac{d^2y}{dt^2} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2 y = \omega_n^2 u(t) \quad (3.7)$$

where:

- ω_n = natural frequency (rad/s)
- ζ = damping ratio (dimensionless)

The **characteristic equation** is:

$$\lambda^2 + 2\zeta\omega_n \lambda + \omega_n^2 = 0 \quad (3.8)$$

with roots:

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} \quad (3.9)$$

Cases based on damping ratio

1. **Overdamped** ($\zeta > 1$): Two distinct real roots. Response: $y(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$
2. **Critically Damped** ($\zeta = 1$): Repeated real root $\lambda = -\omega_n$. Response: $y(t) = (C_1 + C_2 t) e^{-\omega_n t}$
3. **Underdamped** ($0 < \zeta < 1$): Complex conjugate roots $\lambda = -\zeta\omega_n \pm j\omega_d$ where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Response:

$$y(t) = e^{-\zeta\omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t) \quad (3.10)$$
4. **Undamped** ($\zeta = 0$): Pure imaginary roots. Response: $y(t) = C_1 \cos \omega_n t + C_2 \sin \omega_n t$

Example: Mass-Spring-Damper System

Consider a mass-spring-damper system with $m = 2$ kg, $c = 4$ N·s/m, and $k = 10$ N/m. The equation of motion is:

$$2\ddot{x} + 4\dot{x} + 10x = 0 \quad (3.11)$$

Step 1: Write in standard form by dividing by m :

$$\ddot{x} + 2\dot{x} + 5x = 0 \quad (3.12)$$

Step 2: Identify parameters. Comparing with $\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$:

- $\omega_n^2 = 5 \Rightarrow \omega_n = \sqrt{5} \approx 2.24$ rad/s
- $2\zeta\omega_n = 2 \Rightarrow \zeta = 1/\omega_n = 1/\sqrt{5} \approx 0.447$

Step 3: Since $\zeta < 1$, the system is **underdamped**. The damped frequency is:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = \sqrt{5} \sqrt{1 - 0.2} = \sqrt{4} = 2 \text{ rad/s} \quad (3.13)$$

Step 4: The general solution is:

$$x(t) = e^{-t} (C_1 \cos 2t + C_2 \sin 2t) \quad (3.14)$$

With initial conditions $x(0) = 1$ m, $\dot{x}(0) = 0$:

- From $x(0) = 1$: $C_1 = 1$
- From $\dot{x}(0) = 0$: $-C_1 + 2C_2 = 0 \Rightarrow C_2 = 0.5$

Final answer:

$$x(t) = e^{-t} (\cos 2t + 0.5 \sin 2t) \text{ m} \quad (3.15)$$

Example: Overdamped Second-Order System

Solve $\ddot{y} + 5\dot{y} + 4y = 0$ with $y(0) = 1$, $\dot{y}(0) = 0$.

Step 1: Find the characteristic equation:

$$\lambda^2 + 5\lambda + 4 = 0 \quad (3.16)$$

Step 2: Factor to find roots:

$$(\lambda + 1)(\lambda + 4) = 0 \Rightarrow \lambda_1 = -1, \quad \lambda_2 = -4 \quad (3.17)$$

Since both roots are real and distinct (and negative), the system is **overdamped and stable**.

Step 3: Write the general solution:

$$y(t) = C_1 e^{-t} + C_2 e^{-4t} \quad (3.18)$$

Step 4: Apply initial conditions:

$$y(0) = C_1 + C_2 = 1 \quad (3.19)$$

$$\dot{y}(0) = -C_1 - 4C_2 = 0 \quad (3.20)$$

Solving: $C_1 = 4/3$, $C_2 = -1/3$.

Final answer:

$$y(t) = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t} \quad (3.21)$$

Note: The response is dominated by the slower mode e^{-t} as t increases.

Example: Critically Damped System

Solve $\ddot{y} + 6\dot{y} + 9y = 0$ with $y(0) = 2$, $\dot{y}(0) = 3$.

Step 1: Characteristic equation:

$$\lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0 \quad (3.22)$$

Step 2: Repeated root $\lambda = -3$ (critically damped, $\zeta = 1$, $\omega_n = 3$).

Step 3: General solution for repeated roots:

$$y(t) = (C_1 + C_2 t)e^{-3t} \quad (3.23)$$

Step 4: Apply initial conditions:

$$y(0) = C_1 = 2 \quad (3.24)$$

$$\dot{y}(t) = C_2 e^{-3t} - 3(C_1 + C_2 t)e^{-3t} \quad (3.25)$$

$$\dot{y}(0) = C_2 - 3C_1 = 3 \Rightarrow C_2 = 9 \quad (3.26)$$

Final answer:

$$y(t) = (2 + 9t)e^{-3t} \quad (3.27)$$

This represents the fastest decay without oscillation.

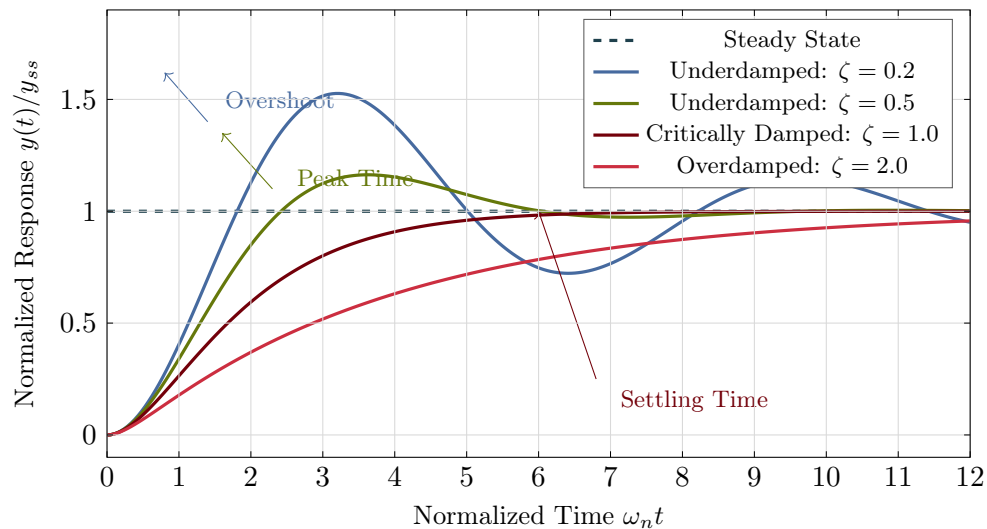


Figure 3.2: Step response of second-order systems for different damping ratios. The response shows how damping affects the transient behavior: underdamped ($\zeta < 1$) exhibits overshoot and oscillation, critically damped ($\zeta = 1$) provides fast response without overshoot, and overdamped ($\zeta > 1$) is slow but smooth. Key features include overshoot (Atlantic blue), peak time (Horseshoe green), and settling time (Garnet red) annotations. Dashed line shows steady-state value.

Common Pitfall: Sign Errors in Characteristic Equation

When solving the characteristic equation, be careful with signs:

- The ODE $\ddot{y} + a\dot{y} + by = 0$ gives characteristic equation $\lambda^2 + a\lambda + b = 0$
- Stable systems require $a > 0$ and $b > 0$ (all coefficients positive)
- A negative coefficient indicates instability

Example: $\ddot{y} - 2\dot{y} + y = 0$ has $\lambda = 1$ (repeated), which is **unstable** despite the positive coefficient of y .

3.1.3 Higher-Order Linear ODEs

The general n th-order linear ODE with constant coefficients:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = f(t) \quad (3.28)$$

The solution consists of:

- **Homogeneous solution** $y_h(t)$: From the characteristic equation roots

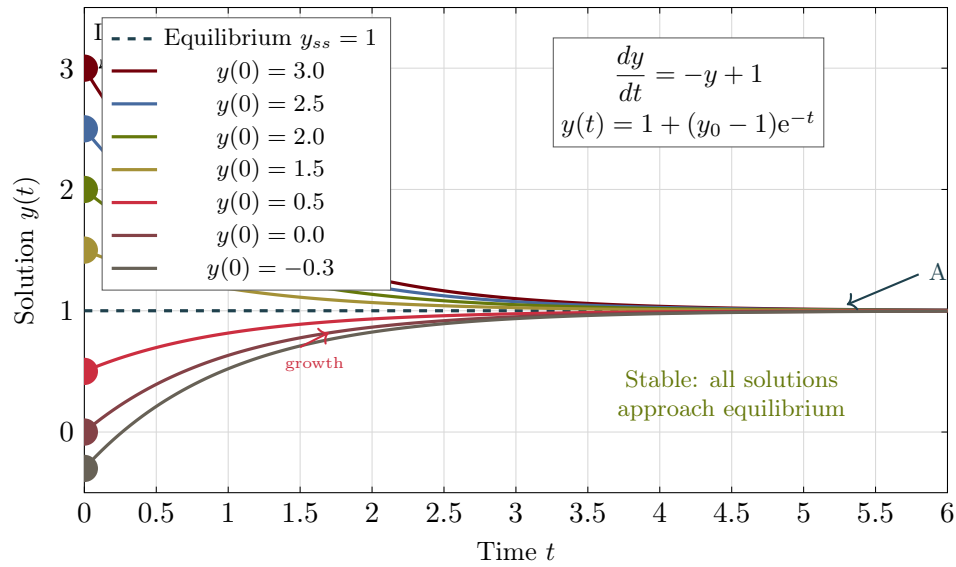


Figure 3.3: Family of solutions to the first-order ODE $\frac{dy}{dt} = -y + 1$ for different initial conditions. The general solution $y(t) = 1 + (y_0 - 1)e^{-t}$ demonstrates that all trajectories converge exponentially to the stable equilibrium $y_{ss} = 1$ (dashed Congaree line). Solutions starting above the equilibrium decay downward, while solutions starting below the equilibrium grow upward. This convergence behavior, regardless of initial condition, characterizes asymptotic stability: the equilibrium is globally attractive. The rate of convergence is determined by the eigenvalue $\lambda = -1$ (time constant $\tau = 1$).

- **Particular solution** $y_p(t)$: Depends on the forcing function $f(t)$

The complete solution is $y(t) = y_h(t) + y_p(t)$.

Example: Converting 3rd Order ODE to State-Space Form

Convert the following third-order ODE to state-space form:

$$\ddot{y} + 6\ddot{y} + 11\dot{y} + 6y = u(t) \quad (3.29)$$

Step 1: Define state variables. Choose:

$$x_1 = y \quad (3.30)$$

$$x_2 = \dot{y} = \dot{x}_1 \quad (3.31)$$

$$x_3 = \ddot{y} = \dot{x}_2 \quad (3.32)$$

Step 2: Write the first-order equations. From the original ODE:

$$\dot{x}_3 = \ddot{y} = -6\ddot{y} - 11\dot{y} - 6y + u = -6x_3 - 11x_2 - 6x_1 + u \quad (3.33)$$

Step 3: Assemble into matrix form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad (3.34)$$

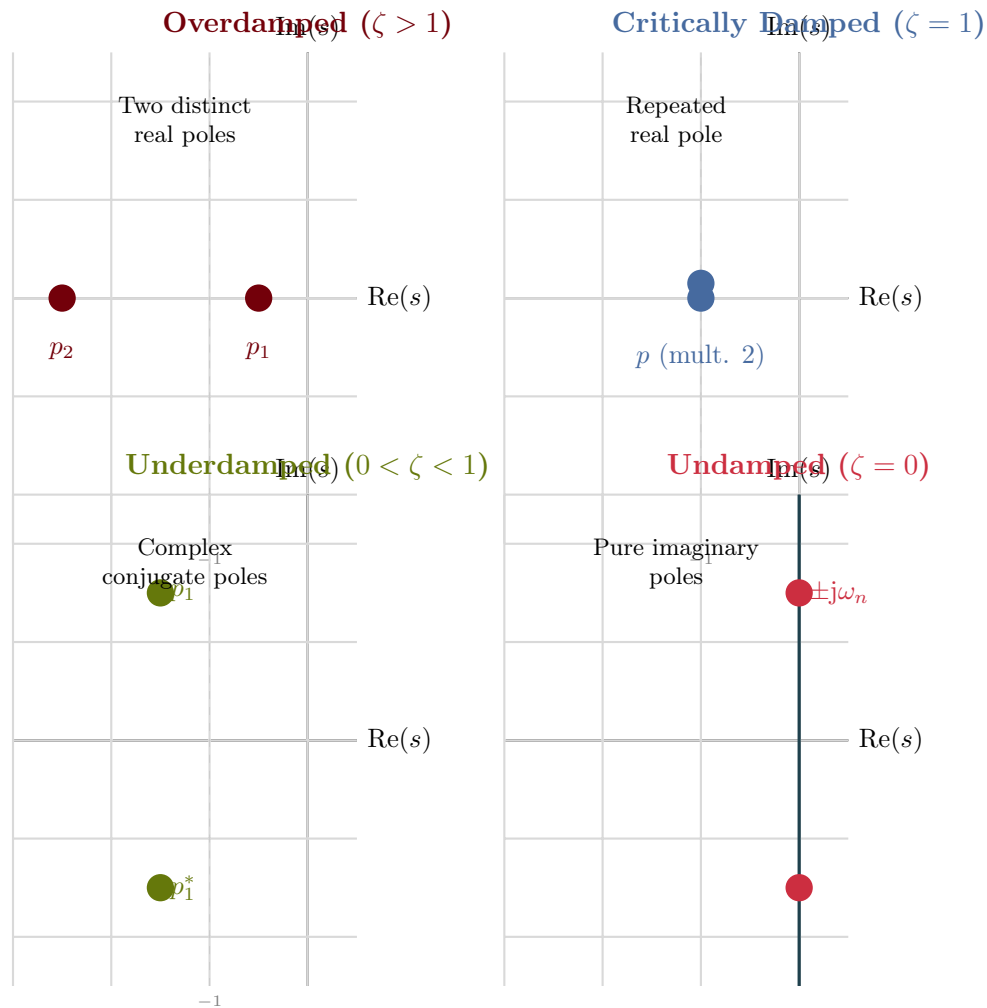


Figure 3.4: Pole locations in the s -plane for second-order systems under different damping conditions. Each subplot shows the characteristic pole configuration: overdamped (Garnet, two real poles on negative axis), critically damped (Atlantic, repeated real pole), underdamped (Horseshoe, complex conjugate pair), and undamped (UofSCRose, pure imaginary pair on the imaginary axis). The stability region (left half-plane) ensures all poles are in the left half-plane for stable responses.

Step 4: Output equation (if y is the output):

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{C}\mathbf{x} \quad (3.35)$$

This is the **controllable canonical form**. The characteristic polynomial of $\mathbf{A}$ matches the original ODE: $\det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^3 + 6\lambda^2 + 11\lambda + 6 = (\lambda + 1)(\lambda + 2)(\lambda + 3)$.

State-Space Conversion Recipe

For an n th-order ODE: $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1\dot{y} + a_0y = b_0u$

1. Define states: $x_1 = y, x_2 = \dot{y}, \dots, x_n = y^{(n-1)}$
2. The companion matrix form is:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} \quad (3.36)$$

3. Output: $\mathbf{C} = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}, \mathbf{D} = 0$

3.2 Solution Techniques for Nonhomogeneous ODEs

The complete solution of a nonhomogeneous linear ODE $L[y] = f(t)$ is:

$$y(t) = y_h(t) + y_p(t) \quad (3.37)$$

where $y_h(t)$ is the homogeneous solution (from Section 3) and $y_p(t)$ is a particular solution.

3.2.1 Method of Undetermined Coefficients

This method works when the forcing function $f(t)$ is a polynomial, exponential, sine, cosine, or a combination thereof.

Definition: Method of Undetermined Coefficients

For an ODE with constant coefficients $L[y] = f(t)$:

1. Identify the form of $f(t)$ from the table below
2. Guess a particular solution $y_p(t)$ with undetermined coefficients
3. Substitute $y_p(t)$ into the ODE and solve for the coefficients
4. If any term in y_p is a solution of the homogeneous equation, multiply by t (or t^2 for repeated roots)

Table of Particular Solution Forms

Forcing Function $f(t)$	Particular Solution Form $y_p(t)$
K (constant)	A
Kt^n	$A_nt^n + A_{n-1}t^{n-1} + \cdots + A_1t + A_0$
Ke^{at}	Ae^{at}
$K \cos(\omega t)$ or $K \sin(\omega t)$	$A \cos(\omega t) + B \sin(\omega t)$
$Ke^{at} \cos(\omega t)$ or $Ke^{at} \sin(\omega t)$	$e^{at}(A \cos(\omega t) + B \sin(\omega t))$
$Kt^n e^{at}$	$(A_nt^n + \cdots + A_0)e^{at}$

Modification Rule for Resonance

If any term in the guessed $y_p(t)$ is also a solution of the homogeneous equation:

- Multiply the entire guess by t
- If still a homogeneous solution (repeated roots), multiply by t^2

This accounts for **resonance** conditions.

Example: Undetermined Coefficients with Exponential Forcing

Solve $\ddot{y} + 3\dot{y} + 2y = 10e^{-3t}$ with $y(0) = 0$, $\dot{y}(0) = 0$.

Step 1: Find homogeneous solution. Characteristic equation: $\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0$

$$y_h(t) = C_1 e^{-t} + C_2 e^{-2t} \quad (3.38)$$

Step 2: Guess particular solution. Since $f(t) = 10e^{-3t}$ and e^{-3t} is not a homogeneous solution:

$$y_p(t) = Ae^{-3t} \quad (3.39)$$

Step 3: Substitute into ODE:

$$\dot{y}_p = -3Ae^{-3t}, \quad \ddot{y}_p = 9Ae^{-3t} \quad (3.40)$$

$$9Ae^{-3t} + 3(-3Ae^{-3t}) + 2(Ae^{-3t}) = 10e^{-3t} \quad (3.41)$$

$$(9A - 9A + 2A)e^{-3t} = 10e^{-3t} \quad (3.42)$$

$$2A = 10 \Rightarrow A = 5 \quad (3.43)$$

Step 4: General solution: $y(t) = C_1 e^{-t} + C_2 e^{-2t} + 5e^{-3t}$

Step 5: Apply initial conditions:

$$y(0) = C_1 + C_2 + 5 = 0 \quad (3.44)$$

$$\dot{y}(0) = -C_1 - 2C_2 - 15 = 0 \quad (3.45)$$

Solving: $C_1 = -20$, $C_2 = 15$.

Final answer:

$$y(t) = -20e^{-t} + 15e^{-2t} + 5e^{-3t} \quad (3.46)$$

Example: Undetermined Coefficients with Sinusoidal Forcing

Find the particular solution of $\ddot{y} + 4y = 3 \cos(2t)$.

Step 1: Check homogeneous solutions. Characteristic equation: $\lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2j$

$$y_h(t) = C_1 \cos(2t) + C_2 \sin(2t) \quad (3.47)$$

Step 2: Initial guess would be $A \cos(2t) + B \sin(2t)$, but this IS a homogeneous solution. Apply modification rule:

$$y_p(t) = t(A \cos(2t) + B \sin(2t)) \quad (3.48)$$

Step 3: Compute derivatives:

$$\dot{y}_p = A \cos(2t) + B \sin(2t) + t(-2A \sin(2t) + 2B \cos(2t)) \quad (3.49)$$

$$\ddot{y}_p = -4A \sin(2t) + 4B \cos(2t) - 4t(A \cos(2t) + B \sin(2t)) \quad (3.50)$$

Step 4: Substitute into $\ddot{y}_p + 4y_p = 3 \cos(2t)$:

$$-4A \sin(2t) + 4B \cos(2t) = 3 \cos(2t) \quad (3.51)$$

Comparing coefficients: $B = 3/4$, $A = 0$.

Final answer:

$$y_p(t) = \frac{3}{4}t \sin(2t) \quad (3.52)$$

Note: The factor of t indicates **resonance**—the response grows linearly with time.

3.2.2 Variation of Parameters

The method of variation of parameters provides a general formula that works for *any* forcing function $f(t)$, not just the special forms required for undetermined coefficients.

Variation of Parameters Formula (Second-Order)

For $\ddot{y} + p(t)\dot{y} + q(t)y = f(t)$ with homogeneous solutions $y_1(t)$ and $y_2(t)$:

$$y_p(t) = -y_1(t) \int \frac{y_2(t)f(t)}{W(t)} dt + y_2(t) \int \frac{y_1(t)f(t)}{W(t)} dt \quad (3.53)$$

where $W(t) = y_1\dot{y}_2 - \dot{y}_1y_2$ is the **Wronskian**.

For constant-coefficient equations, the Wronskian is constant (or a simple exponential), making the integrals more tractable.

When to Use Each Method

- **Undetermined coefficients:** Use when forcing is polynomial, exponential, sine/cosine, or products thereof. Faster and simpler.
- **Variation of parameters:** Use when forcing is more complex (e.g., $\tan t$, $\sec t$, arbitrary functions). Always works but integrals may be difficult.

- **Laplace transforms:** Often the easiest approach for control systems (Chapter ??). Handles initial conditions automatically.

3.3 Phase Plane Analysis

The phase plane provides a geometric visualization of second-order system behavior without explicitly solving the differential equation.

Definition: Phase Portrait

For a second-order system written as two first-order equations:

$$\dot{x}_1 = f_1(x_1, x_2) \quad (3.54)$$

$$\dot{x}_2 = f_2(x_1, x_2) \quad (3.55)$$

The **phase portrait** is the collection of all trajectories in the (x_1, x_2) plane, showing how the system state evolves from different initial conditions.

For linear systems $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, the eigenvalues of $\mathbf{A}$ determine the type of equilibrium:

Eigenvalues	Equilibrium Type	Stability
$\lambda_1, \lambda_2 < 0$ (real, distinct)	Stable node	Asymptotically stable
$\lambda_1, \lambda_2 > 0$ (real, distinct)	Unstable node	Unstable
$\lambda_1 < 0 < \lambda_2$ (real, opposite signs)	Saddle point	Unstable
$\lambda = \sigma \pm j\omega, \sigma < 0$	Stable focus (spiral)	Asymptotically stable
$\lambda = \sigma \pm j\omega, \sigma > 0$	Unstable focus (spiral)	Unstable
$\lambda = \pm j\omega$ (pure imaginary)	Center	Marginally stable

3.4 Stability Analysis

Stability is a fundamental property in control systems. A system is stable if its response remains bounded and eventually settles to equilibrium.

3.4.1 Definitions of Stability

Definition: Lyapunov Stability

An equilibrium point $\mathbf{x}_e$ is **Lyapunov stable** (or stable in the sense of Lyapunov) if, for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\|\mathbf{x}(0) - \mathbf{x}_e\| < \delta \Rightarrow \|\mathbf{x}(t) - \mathbf{x}_e\| < \epsilon \quad \forall t \geq 0 \quad (3.56)$$

In other words, trajectories starting close to equilibrium remain close.

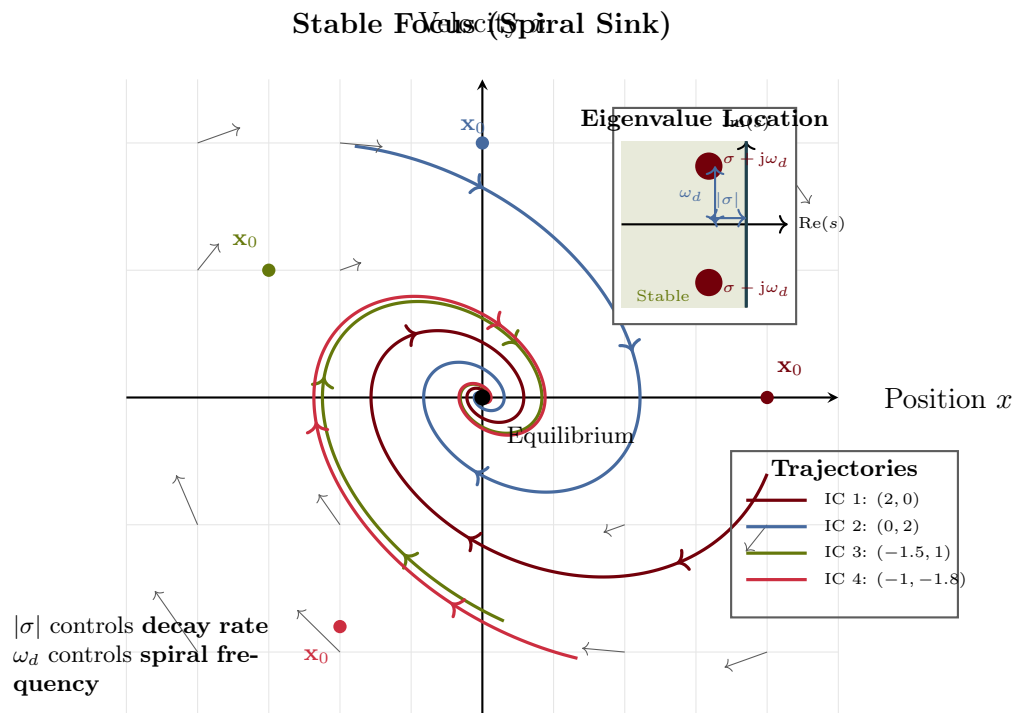


Figure 3.5: Phase portrait for a stable second-order system (stable focus, $\zeta = 0.3$, $\omega_n = 1$). Trajectories in the state space $(x, \dot{x})$ spiral inward toward the equilibrium at the origin, demonstrating asymptotic stability. The spiral pattern results from complex conjugate eigenvalues $\lambda = \sigma \pm j\omega_d$ with negative real part ($\sigma < 0$). The inset shows eigenvalue locations in the s -plane: the real part σ determines the decay rate (how quickly spirals shrink), while the imaginary part ω_d determines the oscillation frequency (how tightly spirals wind). All trajectories from different initial conditions converge to the stable equilibrium.

Definition: Asymptotic Stability

An equilibrium is **asymptotically stable** if it is Lyapunov stable AND:

$$\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{x}_e \quad (3.57)$$

for all initial conditions in some neighborhood of $\mathbf{x}_e$. The system not only stays close but converges to equilibrium.

Definition: BIBO Stability

A system is **Bounded-Input Bounded-Output (BIBO) stable** if every bounded input produces a bounded output:

$$|u(t)| \leq M_u < \infty \quad \forall t \geq 0 \Rightarrow |y(t)| \leq M_y < \infty \quad \forall t \geq 0 \quad (3.58)$$

This is the most common stability notion for input-output (transfer function) descriptions.

Stability Criterion for LTI Systems

For a linear time-invariant (LTI) system, the following are equivalent:

1. The system is asymptotically stable
2. All eigenvalues of $\mathbf{A}$ have negative real parts: $\text{Re}(\lambda_i) < 0$
3. All poles of the transfer function lie in the open left half-plane
4. The impulse response decays to zero: $\lim_{t \rightarrow \infty} h(t) = 0$

For BIBO stability, the additional requirement is that there are no pole-zero cancellations in the right half-plane.

3.4.2 Pole Locations and System Response

The location of poles (eigenvalues) in the complex plane directly determines system behavior:

- **Real part** $\sigma = \text{Re}(\lambda)$: Determines stability and decay/growth rate
 - $\sigma < 0$: Decaying response (stable)
 - $\sigma = 0$: Sustained oscillation or constant (marginally stable)
 - $\sigma > 0$: Growing response (unstable)
- **Imaginary part** $\omega = \text{Im}(\lambda)$: Determines oscillation frequency
 - $\omega = 0$: No oscillation (real pole)
 - $\omega \neq 0$: Oscillation at frequency $|\omega|$ rad/s
- **Distance from origin** $|\lambda|$: Determines speed of response
 - Larger $|\lambda|$: Faster response
 - Smaller $|\lambda|$: Slower response (dominant pole)

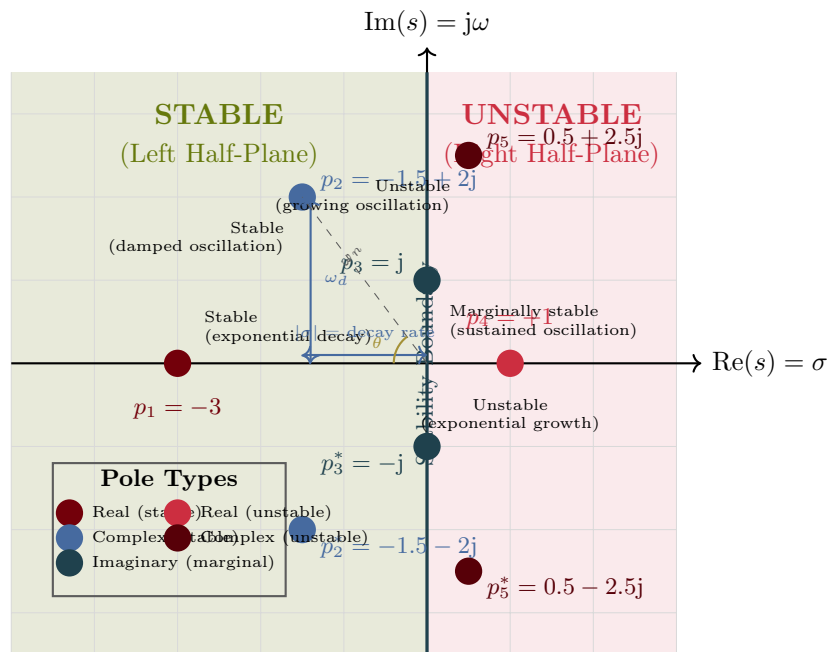


Figure 3.6: Stability regions in the complex s -plane. The left half-plane (LHP, shaded green) contains stable poles where $\text{Re}(s) < 0$, producing decaying responses. The right half-plane (RHP, shaded pink) contains unstable poles where $\text{Re}(s) > 0$, producing unbounded responses. The imaginary axis (Congaree, emphasized) is the stability boundary: poles on this axis yield marginally stable systems with sustained oscillations. Example poles demonstrate different behaviors: real stable pole (p_1 , exponential decay), complex stable pair (p_2, p_2^* , damped oscillation), imaginary pair (p_3, p_3^* , sustained oscillation), real unstable pole (p_4 , exponential growth), and complex unstable pair (p_5, p_5^* , growing oscillation).

3.4.3 The Routh-Hurwitz Criterion (Preview)

For higher-order systems, directly computing eigenvalues can be difficult. The **Routh-Hurwitz criterion** provides an algebraic test for stability without solving the characteristic equation.

Routh-Hurwitz Stability Criterion

For a characteristic polynomial:

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0 \quad (3.59)$$

Necessary conditions (quick check):

1. All coefficients a_i must be present (no missing terms)
2. All coefficients must have the same sign (typically all positive)

If either condition fails, the system is unstable.

The **sufficient condition** requires constructing the Routh array and checking that all entries in the first column have the same sign. The number of sign changes equals the number of right half-plane poles.

Note: The complete Routh-Hurwitz procedure is covered in detail in the classical control chapters.

Example: Quick Stability Check

Determine if the following characteristic polynomials represent stable systems:

(a) $s^3 + 4s^2 + 6s + 4 = 0$

All coefficients present and positive. Passes necessary conditions. (Routh array confirms stability: roots are $-2, -1 \pm j$.)

(b) $s^3 + 2s^2 + s + 8 = 0$

All coefficients present and positive. Passes necessary conditions. However, Routh array reveals two RHP poles—**unstable**.

(c) $s^4 + 2s^2 + 1 = 0$

Missing s^3 and s^1 terms. Fails necessary condition—**unstable** (or marginally stable with imaginary axis poles).

(d) $s^3 - 2s^2 + s + 4 = 0$

Coefficient of s^2 is negative while others are positive. Sign inconsistency—**unstable**.

Marginal Stability Warning

Systems with poles exactly on the imaginary axis ($\text{Re}(\lambda) = 0$) are **marginally stable**:

- Bounded response to bounded inputs only for some input types
- Sustained oscillations (not decaying, not growing)
- Any perturbation can push the system to instability
- In practice, treat marginally stable systems as potentially unstable

3.5 Systems of First-Order ODEs

Any higher-order ODE can be written as a system of first-order equations. This is the **state-space** representation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (3.60)$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the input vector, and $\mathbf{A}$, $\mathbf{B}$ are system matrices.

The solution is:

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau \quad (3.61)$$

where $e^{\mathbf{A}t}$ is the **matrix exponential** (discussed in Chapter ??).

Connection: Poles

For a linear system, the following are all equivalent characterizations:

Representation	Stability Determined By	Condition
ODE: $a_n y^{(n)} + \cdots + a_0 y = f(t)$	Roots of characteristic polynomial	$\text{Re}(\lambda_i) < 0$
Transfer function: $G(s) = N(s)/D(s)$	Poles (roots of $D(s)$)	$\text{Re}(p_i) < 0$
State-space: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$	Eigenvalues of $\mathbf{A}$	$\text{Re}(\lambda_i) < 0$

All three approaches yield the same stability conclusion for a given system.

Chapter 4

Laplace Transforms

The Laplace transform converts differential equations into algebraic equations, making system analysis much simpler.

4.1 Definition

Definition 0.2: Laplace Transform

The **Laplace transform** of a function $f(t)$ defined for $t \geq 0$ is:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt \quad (4.1)$$

where $s = \sigma + j\omega$ is a complex variable.

The complex variable $s = \sigma + j\omega$ defines a plane called the **s-plane** or **complex frequency plane**. The real part σ determines exponential growth or decay, while the imaginary part ω determines oscillation frequency.

The **inverse Laplace transform** is:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds \quad (4.2)$$

In practice, we use tables and partial fraction expansion rather than direct integration.

Example: Deriving the Laplace Transform of te^{-at} from Definition

Derive the Laplace transform of $f(t) = te^{-at}$ directly from the definition.

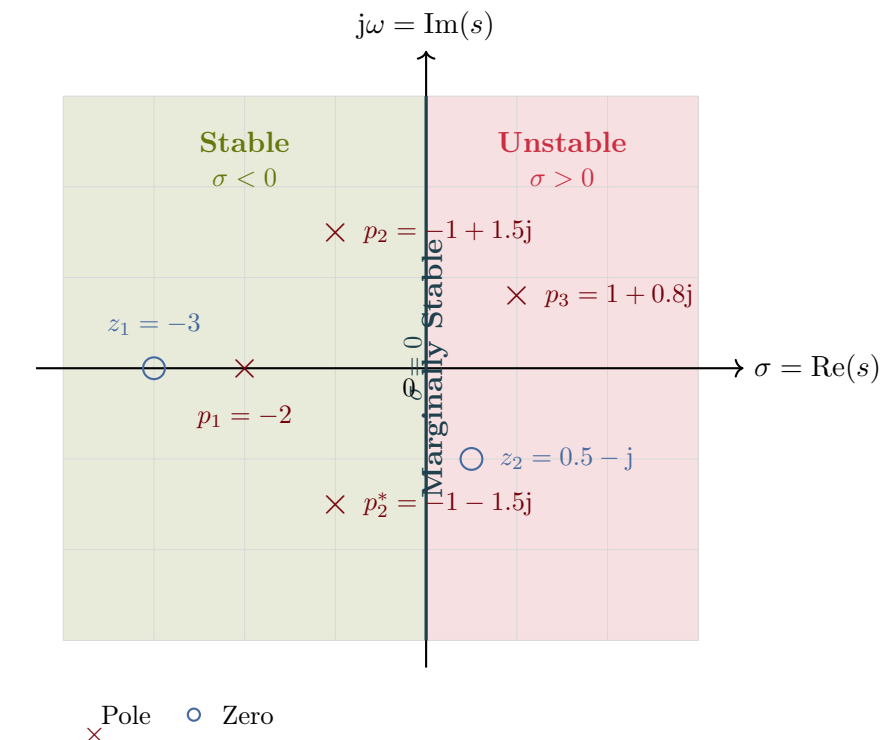
Solution: Applying the definition:

$$F(s) = \int_0^{\infty} te^{-at}e^{-st} dt = \int_0^{\infty} te^{-(s+a)t} dt \quad (4.3)$$

Use integration by parts with $u = t$ and $dv = e^{-(s+a)t}dt$:

$$du = dt \quad (4.4)$$

$$v = \frac{-1}{s+a}e^{-(s+a)t} \quad (4.5)$$



Time Response Characteristics by Pole Location

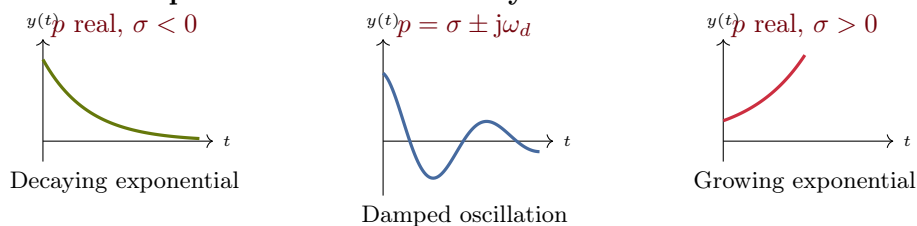


Figure 4.1: The complex s -plane showing stability regions and pole-zero locations. The left half-plane (Horseshoe green shading, $\sigma < 0$) corresponds to stable system responses, while the right half-plane (Rose shading, $\sigma > 0$) produces unstable responses. The imaginary axis (Congaree, $\sigma = 0$) represents marginal stability. Poles are marked with Garnet $\times$ symbols, zeros with Atlantic $\circ$ symbols. Bottom panels show representative time responses: stable real poles produce decaying exponentials, complex conjugate poles in the LHP produce damped oscillations, and RHP poles lead to unbounded growth.

Applying integration by parts:

$$F(s) = \left[\frac{-t}{s+a} e^{-(s+a)t} \right]_0^\infty + \frac{1}{s+a} \int_0^\infty e^{-(s+a)t} dt \quad (4.6)$$

$$= 0 + \frac{1}{s+a} \cdot \frac{1}{s+a} \quad (4.7)$$

$$= \frac{1}{(s+a)^2} \quad (4.8)$$

where we used the fact that $\lim_{t \rightarrow \infty} te^{-(s+a)t} = 0$ for $\text{Re}(s) > -a$.

Therefore:

$$\boxed{\mathcal{L}\{te^{-at}\} = \frac{1}{(s+a)^2}} \quad (4.9)$$

4.2 Region of Convergence

The Laplace transform integral does not converge for all values of s . The set of s values for which the integral converges is called the **Region of Convergence (ROC)**.

Definition: Region of Convergence

The **Region of Convergence (ROC)** of a Laplace transform $F(s)$ is the set of all complex values s for which the integral $\int_0^\infty f(t)e^{-st} dt$ converges (exists and is finite).

For most signals encountered in control systems, the ROC is a half-plane of the form $\text{Re}(s) > \sigma_0$, where σ_0 is called the **abscissa of convergence**.

ROC and Causality

The ROC determines the causality of the corresponding time-domain signal:

- **Right-sided signals** (causal, $f(t) = 0$ for $t < 0$): ROC is a right half-plane $\text{Re}(s) > \sigma_0$
- **Left-sided signals** (anti-causal): ROC is a left half-plane $\text{Re}(s) < \sigma_0$
- **Two-sided signals**: ROC is a vertical strip $\sigma_1 < \text{Re}(s) < \sigma_2$

Important Note

In control systems, we **always assume causal (right-sided) signals**. This means we work with ROCs that are right half-planes extending to $\text{Re}(s) \rightarrow +\infty$. This assumption is implicit in all transform pair tables and partial fraction methods.

For example, the transform pair $e^{-at} \leftrightarrow \frac{1}{s+a}$ assumes causality. The same $F(s) = \frac{1}{s+a}$ with a left half-plane ROC would correspond to $-e^{-at}u(-t)$, an anti-causal signal.

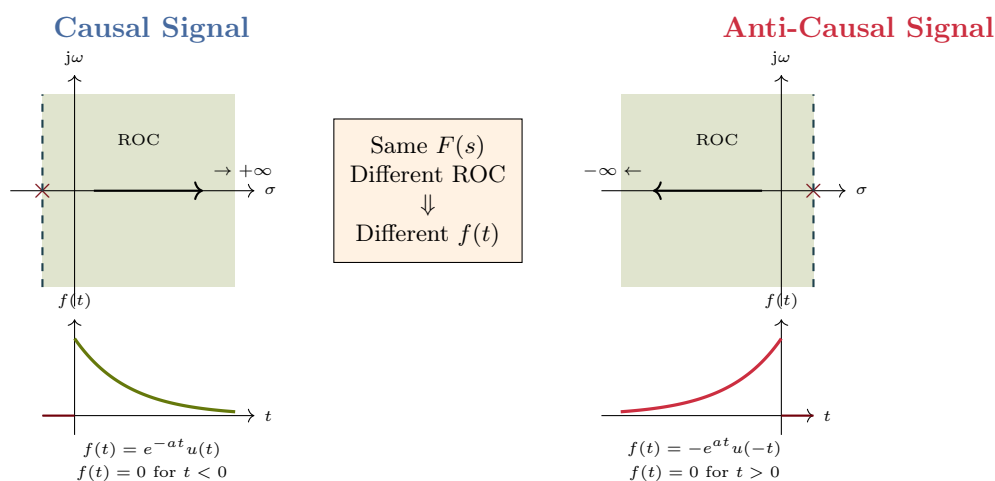
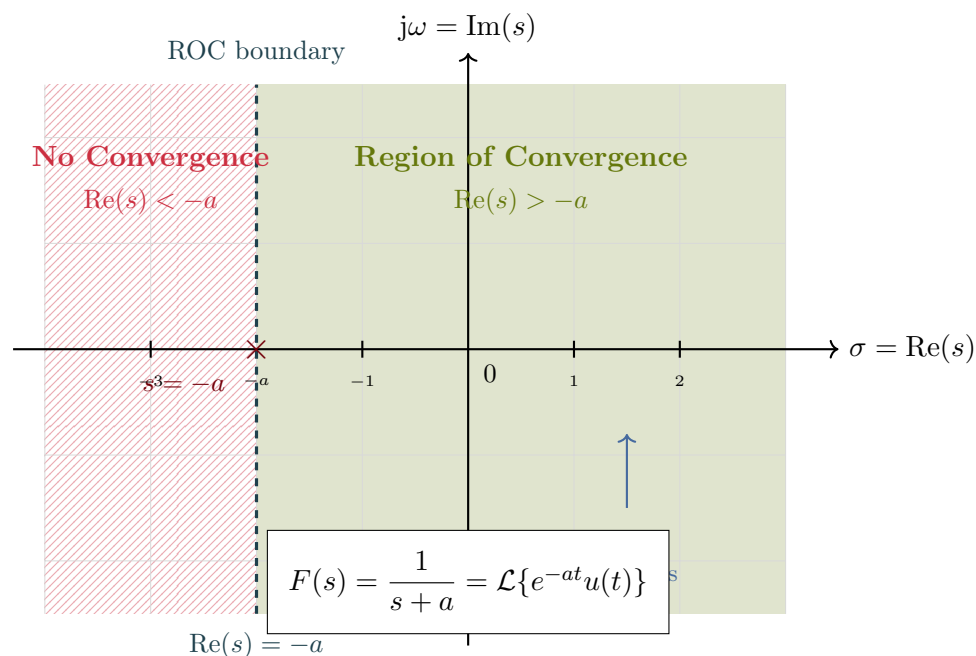


Figure 4.2: Region of Convergence (ROC) for the Laplace transform $F(s) = 1/(s + a)$. The ROC (Horseshoe green shading) is the half-plane $\text{Re}(s) > -a$, to the right of the pole at $s = -a$ (Garnet $\times$). The hatched region shows where the Laplace integral does not converge. The ROC determines whether the inverse transform yields a causal or anti-causal time-domain signal: ROC extending to $+\infty$ corresponds to the causal signal $e^{-at}u(t)$, while ROC extending to $-\infty$ yields the anti-causal signal $-e^{at}u(-t)$. For control systems, we typically assume causal signals, so the ROC extends rightward from the rightmost pole.

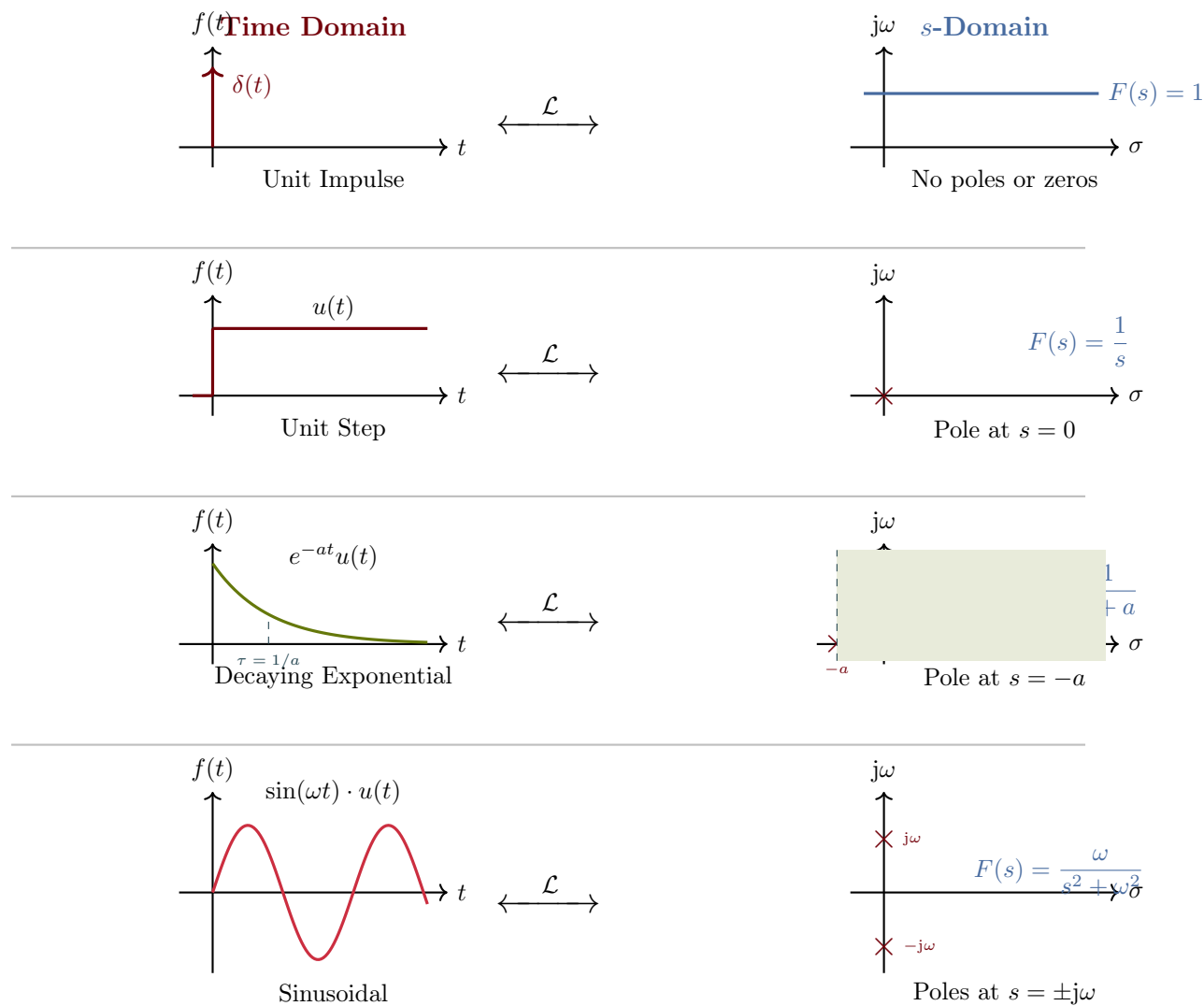


Figure 4.3: Visual comparison of fundamental Laplace transform pairs showing the correspondence between time-domain signals (left, Garnet/Horseshoe/Rose) and their s -domain representations (right, Atlantic). From top to bottom: (1) the unit impulse $\delta(t)$ transforms to constant 1 with no poles; (2) the unit step $u(t)$ transforms to $1/s$ with a pole at the origin; (3) the decaying exponential $e^{-at}u(t)$ transforms to $1/(s+a)$ with a pole at $s = -a$ (the shaded region shows the ROC); (4) the sine wave $\sin(\omega t)u(t)$ transforms to $\omega/(s^2 + \omega^2)$ with complex conjugate poles on the imaginary axis. Pole locations directly determine the time-domain behavior.

Signal Name	$f(t)$ for $t \geq 0$	$F(s)$
Impulse	$\delta(t)$	1
Unit step	$u(t) = 1$	$\frac{1}{s}$
Ramp	t	$\frac{1}{s^2}$
Parabolic	t^2	$\frac{2}{s^3}$
Power	t^n	$\frac{n!}{s^{n+1}}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Decaying ramp	te^{-at}	$\frac{1}{(s+a)^2}$
Decaying power	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
Sine	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
Cosine	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
Damped sine	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
Damped cosine	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Table 4.1: Common Laplace transform pairs.

4.3 Common Laplace Transform Pairs

4.4 Laplace Transform Properties

Property	Time Domain	s -Domain
Linearity	$af(t) + bg(t)$	$aF(s) + bG(s)$
Differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
Second derivative	$\frac{d^2f}{dt^2}$	$s^2F(s) - sf(0^-) - f'(0^-)$
n th derivative	$\frac{d^nf}{dt^n}$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^-)$
Integration	$\int_0^t f(\tau) d\tau$	$\frac{F(s)}{s}$
Time delay	$f(t - \tau)u(t - \tau)$	$e^{-\tau s}F(s)$
Frequency shift	$e^{-at}f(t)$	$F(s + a)$
Time scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Multiplication by t	$tf(t)$	$-\frac{dF}{ds}$
Convolution	$f(t) * g(t)$	$F(s)G(s)$
Initial value	$\lim_{t \rightarrow 0^+} f(t)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} sF(s)$

Table 4.2: Laplace transform properties.

Example: Using the Differentiation Property to Find $\mathcal{L}\{t^2\}$

Use the differentiation property of Laplace transforms to find $\mathcal{L}\{t^2\}$.

Solution: The “multiplication by t ” property states:

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds} \quad (4.10)$$

We know that $\mathcal{L}\{t\} = \frac{1}{s^2}$.

Apply the property to find $\mathcal{L}\{t \cdot t\} = \mathcal{L}\{t^2\}$:

$$\mathcal{L}\{t^2\} = \mathcal{L}\{t \cdot t\} = -\frac{d}{ds} \left(\frac{1}{s^2} \right) \quad (4.11)$$

$$= - \left(\frac{-2}{s^3} \right) \quad (4.12)$$

$$= \frac{2}{s^3} \quad (4.13)$$

Therefore:

$$\boxed{\mathcal{L}\{t^2\} = \frac{2}{s^3}} \quad (4.14)$$

This pattern generalizes: applying the property n times gives $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$.

Limitations of the Final Value Theorem

The **Final Value Theorem** only applies if all poles of $sF(s)$ are in the left half-plane (LHP). If any poles are on the imaginary axis or in the right half-plane, the theorem gives incorrect results.

Examples of invalid applications:

- $F(s) = \frac{\omega}{s^2 + \omega^2}$ (sine wave): Poles at $s = \pm j\omega$ on imaginary axis. FVT would incorrectly give 0.
- $F(s) = \frac{1}{s-1}$ (exponentially growing): Pole at $s = 1$ in RHP. FVT would incorrectly give -1 .
- $F(s) = \frac{1}{s^2}$ (ramp): Double pole at $s = 0$. Limit does not exist.

Example: Initial and Final Value Theorems

For the transfer function $F(s) = \frac{10(s+2)}{s(s+1)(s+5)}$, find the initial and final values of $f(t)$.

Solution:

Initial Value Theorem:

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} s \cdot \frac{10(s+2)}{s(s+1)(s+5)} \quad (4.15)$$

$$= \lim_{s \rightarrow \infty} \frac{10(s+2)}{(s+1)(s+5)} \quad (4.16)$$

$$= \lim_{s \rightarrow \infty} \frac{10s+20}{s^2+6s+5} \quad (4.17)$$

$$= \lim_{s \rightarrow \infty} \frac{10/s+20/s^2}{1+6/s+5/s^2} = 0 \quad (4.18)$$

Final Value Theorem: First, check validity: the poles of $sF(s)$ are at $s = -1$ and $s = -5$,

both in the LHP. The theorem is valid.

$$f(\infty) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} s \cdot \frac{10(s+2)}{s(s+1)(s+5)} \quad (4.19)$$

$$= \lim_{s \rightarrow 0} \frac{10(s+2)}{(s+1)(s+5)} \quad (4.20)$$

$$= \frac{10(0+2)}{(0+1)(0+5)} = \frac{20}{5} = 4 \quad (4.21)$$

Therefore:

$$\boxed{f(0^+) = 0, \quad f(\infty) = 4} \quad (4.22)$$

4.5 Partial Fraction Expansion

To find the inverse Laplace transform, we decompose rational functions into simpler terms.

Distinct Real Poles

$$\frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)} = \frac{A_1}{s+p_1} + \frac{A_2}{s+p_2} + \cdots + \frac{A_n}{s+p_n} \quad (4.23)$$

The residue A_i is found by:

$$A_i = \lim_{s \rightarrow -p_i} (s+p_i)F(s) \quad (4.24)$$

Repeated Poles

For a pole of multiplicity m at $s = -p$:

$$\frac{N(s)}{(s+p)^m} = \frac{A_1}{s+p} + \frac{A_2}{(s+p)^2} + \cdots + \frac{A_m}{(s+p)^m} \quad (4.25)$$

The coefficients are found by:

$$A_m = \lim_{s \rightarrow -p} (s+p)^m F(s) \quad (4.26)$$

$$A_{m-1} = \lim_{s \rightarrow -p} \frac{d}{ds} [(s+p)^m F(s)] \quad (4.27)$$

$$A_{m-k} = \lim_{s \rightarrow -p} \frac{1}{k!} \frac{d^k}{ds^k} [(s+p)^m F(s)] \quad (4.28)$$

Complex Conjugate Poles

For poles at $s = -\sigma \pm j\omega$:

$$\frac{N(s)}{(s+\sigma)^2 + \omega^2} = \frac{As+B}{(s+\sigma)^2 + \omega^2} \quad (4.29)$$

Example 0.3: Partial Fraction Expansion with Distinct Real Poles

Find the inverse Laplace transform of:

$$F(s) = \frac{2s + 3}{(s + 1)(s + 2)} \quad (4.30)$$

Solution:

$$F(s) = \frac{A}{s + 1} + \frac{B}{s + 2} \quad (4.31)$$

Residues:

$$A = \lim_{s \rightarrow -1} (s + 1) \cdot \frac{2s + 3}{(s + 1)(s + 2)} = \frac{2(-1) + 3}{-1 + 2} = 1 \quad (4.32)$$

$$B = \lim_{s \rightarrow -2} (s + 2) \cdot \frac{2s + 3}{(s + 1)(s + 2)} = \frac{2(-2) + 3}{-2 + 1} = 1 \quad (4.33)$$

Therefore:

$$f(t) = e^{-t} + e^{-2t}, \quad t \geq 0 \quad (4.34)$$

Example: Partial Fraction Expansion with Complex Poles

Find the inverse Laplace transform of:

$$F(s) = \frac{s + 3}{s^2 + 2s + 5} \quad (4.35)$$

Solution: First, identify the poles by factoring the denominator:

$$s^2 + 2s + 5 = (s + 1)^2 + 4 \quad (4.36)$$

The poles are at $s = -1 \pm 2j$ (complex conjugates with $\sigma = 1$, $\omega = 2$).

Rewrite the numerator to match standard forms. Complete the square in the numerator relative to $(s + 1)$:

$$s + 3 = (s + 1) + 2 \quad (4.37)$$

Therefore:

$$F(s) = \frac{(s + 1) + 2}{(s + 1)^2 + 4} \quad (4.38)$$

$$= \frac{s + 1}{(s + 1)^2 + 4} + \frac{2}{(s + 1)^2 + 4} \quad (4.39)$$

$$= \frac{s + 1}{(s + 1)^2 + 2^2} + \frac{2}{(s + 1)^2 + 2^2} \quad (4.40)$$

Using transform pairs for damped cosine and damped sine:

- $\frac{s+a}{(s+a)^2+\omega^2} \leftrightarrow e^{-at} \cos \omega t$
- $\frac{\omega}{(s+a)^2+\omega^2} \leftrightarrow e^{-at} \sin \omega t$

Therefore:

$$\boxed{f(t) = e^{-t} \cos 2t + e^{-t} \sin 2t = e^{-t}(\cos 2t + \sin 2t), \quad t \geq 0} \quad (4.41)$$

Alternative form: Using the identity $A \cos \theta + B \sin \theta = \sqrt{A^2 + B^2} \cos(\theta - \phi)$ where $\phi = \arctan(B/A)$:

$$f(t) = \sqrt{2}e^{-t} \cos(2t - \pi/4), \quad t \geq 0 \quad (4.42)$$

4.6 Inverse Laplace Transform Techniques

While the formal inverse Laplace transform involves a complex contour integral, practical methods rely on partial fractions and transform tables.

4.6.1 The Cover-Up Method (Heaviside Expansion)

For distinct poles, the **cover-up method** (also called Heaviside's expansion theorem) provides a quick way to find residues.

Cover-Up Method

For $F(s) = \frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$ with distinct poles, the residue at pole $-p_i$ is:

$$A_i = \left. \frac{N(s)}{(s+p_1)\cdots(s+p_i)\cdots(s+p_n)} \right|_{s=-p_i} \quad (4.43)$$

That is: “cover up” the factor $(s+p_i)$ in the denominator and evaluate the rest at $s = -p_i$.

Example: Cover-Up Method

Find $\mathcal{L}^{-1} \left\{ \frac{5s+7}{(s+1)(s+2)(s+3)} \right\}$.

Solution: Using the cover-up method:

$$A_1 = \left. \frac{5s+7}{(s+2)(s+3)} \right|_{s=-1} = \frac{5(-1)+7}{(-1+2)(-1+3)} = \frac{2}{1 \cdot 2} = 1 \quad (4.44)$$

$$A_2 = \left. \frac{5s+7}{(s+1)(s+3)} \right|_{s=-2} = \frac{5(-2)+7}{(-2+1)(-2+3)} = \frac{-3}{(-1)(1)} = 3 \quad (4.45)$$

$$A_3 = \left. \frac{5s+7}{(s+1)(s+2)} \right|_{s=-3} = \frac{5(-3)+7}{(-3+1)(-3+2)} = \frac{-8}{(-2)(-1)} = -4 \quad (4.46)$$

Therefore:

$$F(s) = \frac{1}{s+1} + \frac{3}{s+2} - \frac{4}{s+3} \quad (4.47)$$

And the inverse transform is:

$$\boxed{f(t) = e^{-t} + 3e^{-2t} - 4e^{-3t}, \quad t \geq 0} \quad (4.48)$$

4.6.2 Completing the Square for Complex Poles

When the denominator has complex poles, completing the square converts the quadratic to a form matching standard transform pairs.

Completing the Square

For a quadratic $s^2 + bs + c$, complete the square:

$$s^2 + bs + c = \left(s + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) = (s + \sigma)^2 + \omega^2 \quad (4.49)$$

where $\sigma = b/2$ and $\omega = \sqrt{c - b^2/4}$ (assuming $c > b^2/4$ for complex poles).

Example: Completing the Square

Find $\mathcal{L}^{-1}\left\{\frac{3}{s^2+4s+13}\right\}$.

Solution: Complete the square in the denominator:

$$s^2 + 4s + 13 = (s + 2)^2 - 4 + 13 = (s + 2)^2 + 9 = (s + 2)^2 + 3^2 \quad (4.50)$$

Rewrite $F(s)$:

$$F(s) = \frac{3}{(s + 2)^2 + 3^2} \quad (4.51)$$

This matches the damped sine form $\frac{\omega}{(s+a)^2+\omega^2} \leftrightarrow e^{-at} \sin \omega t$, but we have 3 in the numerator and need $\omega = 3$.

$$F(s) = \frac{3}{(s + 2)^2 + 3^2} = 1 \cdot \frac{3}{(s + 2)^2 + 3^2} \quad (4.52)$$

Therefore:

$$\boxed{f(t) = e^{-2t} \sin 3t, \quad t \geq 0} \quad (4.53)$$

4.6.3 Using the Frequency Shifting Theorem

The frequency shifting property $\mathcal{L}\{e^{-at}f(t)\} = F(s + a)$ allows us to handle damped signals systematically.

Inverse Frequency Shift

If $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L}^{-1}\{F(s + a)\} = e^{-at}f(t) \quad (4.54)$$

To use this: identify the shift a , find the unshifted inverse transform, then multiply by e^{-at} .

Example: Frequency Shifting

Find $\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^3}\right\}$.

Solution: Recognize this as a shifted version of $\frac{1}{s^3}$:

$$F(s) = \frac{1}{(s+3)^3} = G(s+3) \quad \text{where} \quad G(s) = \frac{1}{s^3} \quad (4.55)$$

The inverse of $G(s) = \frac{1}{s^3}$ is:

$$g(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} = \frac{t^2}{2} \quad (4.56)$$

By the frequency shift theorem:

$$f(t) = e^{-3t} \cdot \frac{t^2}{2} = \frac{t^2 e^{-3t}}{2}, \quad t \geq 0 \quad (4.57)$$

4.7 Solving ODEs with Laplace Transforms

The Laplace transform provides a systematic method for solving linear ODEs with constant coefficients.

Procedure for Solving ODEs with Laplace Transforms

1. Take the Laplace transform of both sides of the ODE
2. Use the differentiation property to handle derivatives, substituting initial conditions
3. Solve algebraically for $Y(s)$
4. Find $y(t)$ by taking the inverse Laplace transform

Key formulas for incorporating initial conditions:

$$\mathcal{L}\{y'\} = sY(s) - y(0) \quad (4.58)$$

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) \quad (4.59)$$

$$\mathcal{L}\{y'''\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0) \quad (4.60)$$

Example: Solving a Second-Order ODE

Solve the initial value problem:

$$y'' + 3y' + 2y = e^{-t}, \quad y(0) = 1, \quad y'(0) = 0 \quad (4.61)$$

Solution:

Step 1: Take the Laplace transform of both sides.

$$\mathcal{L}\{y'' + 3y' + 2y\} = \mathcal{L}\{e^{-t}\} \quad (4.62)$$

Step 2: Apply differentiation property with initial conditions.

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - s - 0 = s^2Y(s) - s \quad (4.63)$$

$$\mathcal{L}\{y'\} = sY(s) - y(0) = sY(s) - 1 \quad (4.64)$$

$$\mathcal{L}\{y\} = Y(s) \quad (4.65)$$

Substituting:

$$[s^2Y(s) - s] + 3[sY(s) - 1] + 2Y(s) = \frac{1}{s+1} \quad (4.66)$$

Step 3: Solve for $Y(s)$.

$$s^2Y(s) - s + 3sY(s) - 3 + 2Y(s) = \frac{1}{s+1} \quad (4.67)$$

$$(s^2 + 3s + 2)Y(s) = \frac{1}{s+1} + s + 3 \quad (4.68)$$

$$(s+1)(s+2)Y(s) = \frac{1}{s+1} + \frac{(s+3)(s+1)}{s+1} \quad (4.69)$$

$$(s+1)(s+2)Y(s) = \frac{1 + (s+3)(s+1)}{s+1} \quad (4.70)$$

$$(s+1)(s+2)Y(s) = \frac{1 + s^2 + 4s + 3}{s+1} = \frac{s^2 + 4s + 4}{s+1} = \frac{(s+2)^2}{s+1} \quad (4.71)$$

Therefore:

$$Y(s) = \frac{(s+2)^2}{(s+1)^2(s+2)} = \frac{s+2}{(s+1)^2} \quad (4.72)$$

Step 4: Find the inverse Laplace transform.

Perform partial fraction expansion:

$$Y(s) = \frac{s+2}{(s+1)^2} = \frac{(s+1)+1}{(s+1)^2} = \frac{1}{s+1} + \frac{1}{(s+1)^2} \quad (4.73)$$

Using transform pairs:

- $\frac{1}{s+1} \leftrightarrow e^{-t}$
- $\frac{1}{(s+1)^2} \leftrightarrow te^{-t}$

Final Answer:

$$\boxed{y(t) = e^{-t} + te^{-t} = (1+t)e^{-t}, \quad t \geq 0} \quad (4.74)$$

Verification: We can verify by checking initial conditions:

- $y(0) = (1+0)e^0 = 1 \checkmark$
- $y'(t) = e^{-t} - (1+t)e^{-t} = -te^{-t}$, so $y'(0) = 0 \checkmark$

4.8 Transfer Functions

Definition 0.3: Transfer Function

The **transfer function** $G(s)$ of a linear time-invariant system is the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming zero initial conditions:

$$G(s) = \frac{Y(s)}{U(s)} \quad (4.75)$$

Transfer functions are typically written as:

$$G(s) = \frac{b_ms^m + b_{m-1}s^{m-1} + \cdots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0} = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - p_j)} \quad (4.76)$$

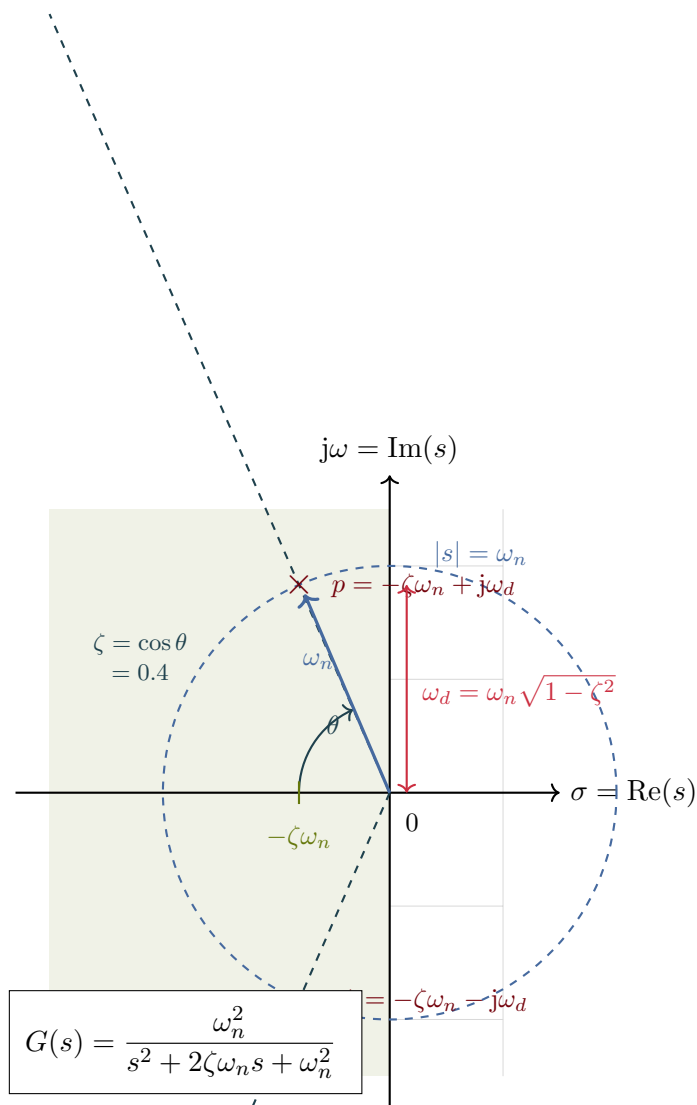
where z_i are the **zeros** and p_j are the **poles** of the transfer function.

Poles and System Behavior

The **poles** of a transfer function determine the system's natural response:

- **Real negative poles** ($s = -a, a > 0$): Decaying exponentials e^{-at}
- **Complex poles** ($s = -\sigma \pm j\omega, \sigma > 0$): Damped oscillations $e^{-\sigma t} \sin(\omega t + \phi)$
- **Poles on imaginary axis** ($s = \pm j\omega$): Sustained oscillations
- **Positive real poles** ($s = a, a > 0$): Unstable exponential growth e^{at}

The **zeros** affect the relative magnitude of different response modes and the transient shape, but do not directly determine stability.

**Pole Locations:**

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

Natural Frequency:

$$\omega_n = |s| = \sqrt{\sigma^2 + \omega_d^2}$$

Damping Ratio:

$$\zeta = \cos \theta = \frac{|\sigma|}{\omega_n}$$

System Response:**Underdamped**

$$0 < \zeta < 1$$

Complex poles

Physical Interpretation: $\sigma = -\zeta\omega_n$: decay rate $\omega_d = \omega_n\sqrt{1-\zeta^2}$: damped frequency $\frac{1}{\zeta\omega_n}$: time constant**Time Response:**

$$y(t) \propto e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$$

Chapter 5

Fourier Transform and Frequency Analysis

5.1 Fourier Series

Any periodic function $f(t)$ with period T can be expressed as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right) \right] \quad (5.1)$$

where:

$$a_0 = \frac{1}{T} \int_0^T f(t) dt \quad (5.2)$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi nt}{T}\right) dt \quad (5.3)$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi nt}{T}\right) dt \quad (5.4)$$

Complex exponential form:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T} \quad (5.5)$$

where $c_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$

5.2 Fourier Transform

For non-periodic functions, the Fourier transform extends the series to continuous frequencies:

Definition 0.4: Fourier Transform

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt \quad (5.6)$$

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{j\omega t} d\omega \quad (5.7)$$

Relationship to Laplace Transform

For causal signals, $F(\omega) = F(s)|_{s=j\omega}$ when the Laplace transform converges on the imaginary axis.

5.3 Frequency Response

For a stable LTI system with transfer function $G(s)$, the **frequency response** is:

$$G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)} \quad (5.8)$$

- $|G(j\omega)|$ = magnitude (gain) at frequency ω
- $\angle G(j\omega)$ = phase shift at frequency ω

For a sinusoidal input $u(t) = A \sin(\omega t)$, the steady-state output is:

$$y_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega)) \quad (5.9)$$

5.4 Bode Plots

Bode plots display magnitude (in dB) and phase versus frequency (on a log scale):

$$|G(j\omega)|_{\text{dB}} = 20 \log_{10} |G(j\omega)| \quad (5.10)$$

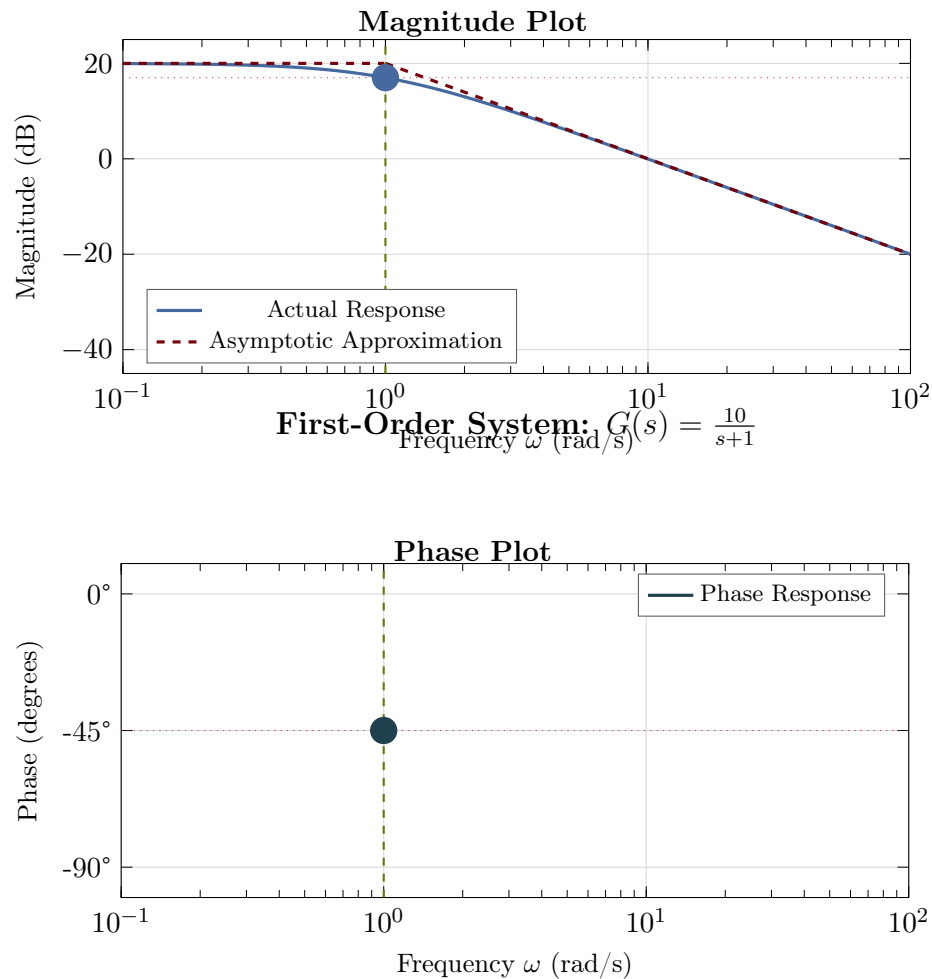


Figure 5.1: Bode plot for a first-order system $G(s) = \frac{10}{s+1}$. The magnitude plot (top, Atlantic blue) shows the steady-state gain of 20 dB below the corner frequency $\omega_c = 1$ rad/s, with a -20 dB/decade slope afterward. The phase plot (bottom, Congaree teal) shows -45° phase lag at the corner frequency. Dashed Garnet lines show asymptotic approximations. The UofSCRose accent color highlights key features: the -3 dB magnitude point and -45° phase point at the corner frequency.

Chapter 6

Linear Algebra

Linear algebra is essential for state-space control, multi-input multi-output (MIMO) systems, and machine learning.

6.1 Vectors and Matrices

Vector Operations

- Addition: $\mathbf{x} + \mathbf{y} = [x_1 + y_1, x_2 + y_2, \dots, x_n + y_n]^\top$
- Scalar multiplication: $\alpha \mathbf{x} = [\alpha x_1, \alpha x_2, \dots, \alpha x_n]^\top$
- Dot product: $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$
- Norm: $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$ (Euclidean norm)

Matrix Operations

- Addition: $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$
- Multiplication: $(\mathbf{AB})_{ij} = \sum_k a_{ik} b_{kj}$
- Transpose: $(\mathbf{A}^\top)_{ij} = a_{ji}$
- Inverse: $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ (if exists)

6.2 Special Matrices

- Identity matrix: $\mathbf{I}$ with ones on diagonal, zeros elsewhere
- Diagonal matrix: $\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$
- Symmetric matrix: $\mathbf{A} = \mathbf{A}^\top$
- Positive definite: $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- Orthogonal matrix: $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$

6.3 Determinant and Rank

The **determinant** of a square matrix:

$$\det(\mathbf{A}) = |\mathbf{A}| = \sum_{j=1}^n (-1)^{1+j} a_{1j} M_{1j} \quad (6.1)$$

where M_{1j} is the minor (determinant of submatrix).

Properties:

- $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$
- $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$
- $\det(\mathbf{A}^\top) = \det(\mathbf{A})$
- $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$

The **rank** is the number of linearly independent rows (or columns):

$$\text{rank}(\mathbf{A}) \leq \min(m, n) \text{ for } \mathbf{A} \in \mathbb{R}^{m \times n} \quad (6.2)$$

6.4 Eigenvalues and Eigenvectors

Definition 0.5: Eigenvalue Problem

For a square matrix $\mathbf{A}$, scalar λ is an **eigenvalue** and non-zero vector $\mathbf{v}$ is the corresponding **eigenvector** if:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (6.3)$$

Eigenvalues are roots of the **characteristic equation**:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \quad (6.4)$$

Properties

- Sum of eigenvalues = $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$
- Product of eigenvalues = $\det(\mathbf{A})$
- Eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i$
- Eigenvalues of $\mathbf{A}^n$ are λ_i^n

Stability and Eigenvalues

A continuous-time system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is:

- **Stable** if all eigenvalues of $\mathbf{A}$ have negative real parts
- **Marginally stable** if eigenvalues have non-positive real parts with simple imaginary-axis eigenvalues

- **Unstable** if any eigenvalue has positive real part

6.5 Matrix Decompositions

Eigendecomposition

For diagonalizable matrix $\mathbf{A}$:

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} \quad (6.5)$$

where $\mathbf{V}$ contains eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Singular Value Decomposition (SVD)

Any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \quad (6.6)$$

where $\mathbf{U}, \mathbf{V}$ are orthogonal and $\mathbf{\Sigma}$ contains singular values.

QR Decomposition

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \quad (6.7)$$

where $\mathbf{Q}$ is orthogonal and $\mathbf{R}$ is upper triangular.

LU Decomposition

$$\mathbf{A} = \mathbf{L}\mathbf{U} \quad (6.8)$$

where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

6.6 Matrix Exponential

The matrix exponential is crucial for solving state-space equations:

Definition 0.6: Matrix Exponential

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!} \quad (6.9)$$

Properties

- $e^0 = \mathbf{I}$
- $\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t}$
- $e^{\mathbf{A}(t_1+t_2)} = e^{\mathbf{A}t_1}e^{\mathbf{A}t_2}$
- $(e^{\mathbf{A}t})^{-1} = e^{-\mathbf{A}t}$

Computation via Laplace

$$e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\} \quad (6.10)$$

Computation via eigendecomposition

If $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$:

$$e^{\mathbf{A}t} = \mathbf{V}e^{\mathbf{\Lambda}t}\mathbf{V}^{-1} = \mathbf{V} \operatorname{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})\mathbf{V}^{-1} \quad (6.11)$$

6.7 Linear Systems of Equations

Solving $\mathbf{A}\mathbf{x} = \mathbf{b}$:

- **Unique solution** if $\mathbf{A}$ is square and invertible: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$
- **No solution** if $\mathbf{b} \notin \operatorname{range}(\mathbf{A})$
- **Infinite solutions** if $\operatorname{null}(\mathbf{A}) \neq \{\mathbf{0}\}$

Least Squares

When $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no exact solution:

$$\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}^\dagger \mathbf{b} \quad (6.12)$$

where $\mathbf{A}^\dagger$ is the Moore-Penrose pseudoinverse.

Chapter 7

Probability and Statistics

Probability theory is essential for stochastic control, Kalman filtering, and machine learning.

7.1 Probability Fundamentals

Axioms of Probability

1. $P(A) \geq 0$ for all events A
2. $P(\Omega) = 1$ where Ω is the sample space
3. For mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (7.1)$$

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (7.2)$$

7.2 Random Variables

Discrete Random Variables

- Probability mass function: $p(x) = P(X = x)$
- Expected value: $E[X] = \sum_x x \cdot p(x)$
- Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$

Continuous Random Variables

- Probability density function: $f(x)$ where $P(a \leq X \leq b) = \int_a^b f(x) dx$
- Expected value: $E[X] = \int_{-\infty}^{\infty} xf(x) dx$
- Cumulative distribution: $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

7.3 Common Distributions

Gaussian (Normal) Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (7.3)$$

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

Multivariate Gaussian

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (7.4)$$

where $\boldsymbol{\mu}$ is the mean vector and $\boldsymbol{\Sigma}$ is the covariance matrix.

Other Important Distributions

- **Uniform:** $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$
- **Exponential:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$
- **Bernoulli:** $P(X = 1) = p$, $P(X = 0) = 1 - p$
- **Poisson:** $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

7.4 Stochastic Processes

A **stochastic process** is a collection of random variables indexed by time: $\{X(t) : t \in T\}$

White Noise

Uncorrelated random sequence with:

- $E[w(t)] = 0$
- $E[w(t)w(\tau)] = \sigma^2 \delta(t - \tau)$

Markov Process

Future states depend only on present state:

$$P(X_{n+1} | X_n, X_{n-1}, \dots, X_0) = P(X_{n+1} | X_n) \quad (7.5)$$

7.5 Estimation Theory

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{ML} = \arg \max_{\theta} \prod_{i=1}^n f(x_i|\theta) = \arg \max_{\theta} \sum_{i=1}^n \log f(x_i|\theta) \quad (7.6)$$

Bayesian Estimation

$$p(\theta|\mathbf{x}) \propto p(\mathbf{x}|\theta)p(\theta) \quad (7.7)$$

(posterior $\propto$ likelihood $\times$ prior)

Minimum Mean Square Error (MMSE)

$$\hat{\theta}_{MMSE} = E[\theta|\mathbf{x}] \quad (7.8)$$

Chapter 8

Optimization

Optimization is central to optimal control, parameter estimation, and machine learning.

8.1 Unconstrained Optimization

Necessary Condition

At a local minimum $\mathbf{x}^*$:

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad (8.1)$$

Sufficient Condition

Also $\mathbf{H}(\mathbf{x}^*) \succ 0$ (positive definite Hessian).

Gradient Descent

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) \quad (8.2)$$

where $\alpha > 0$ is the step size (learning rate).

Newton's Method

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [\mathbf{H}(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k) \quad (8.3)$$

8.2 Constrained Optimization

Equality Constraints

Minimize $f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$

The **Lagrangian**:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) \quad (8.4)$$

Necessary Conditions (KKT)

$$\nabla_{\mathbf{x}} L = \mathbf{0} \quad (8.5)$$

$$\nabla_{\boldsymbol{\lambda}} L = \mathbf{g}(\mathbf{x}) = \mathbf{0} \quad (8.6)$$

Inequality Constraints

Minimize $f(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) \leq \mathbf{0}$

The **Karush-Kuhn-Tucker (KKT) conditions**:

$$\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla h_i(\mathbf{x}^*) = \mathbf{0} \quad (8.7)$$

$$h_i(\mathbf{x}^*) \leq 0 \quad (8.8)$$

$$\mu_i \geq 0 \quad (8.9)$$

$$\mu_i h_i(\mathbf{x}^*) = 0 \quad (\text{complementary slackness}) \quad (8.10)$$

8.3 Convex Optimization

Definition 0.7: Convex Function

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if for all $\mathbf{x}, \mathbf{y}$ and $\theta \in [0, 1]$:

$$f(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y}) \quad (8.11)$$

Key property: Any local minimum of a convex function is a global minimum.

Examples of Convex Problems

- Linear programming: $\min \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{Ax} \leq \mathbf{b}$
- Quadratic programming: $\min \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{Ax} \leq \mathbf{b}$ (if $\mathbf{Q} \succeq 0$)
- Least squares: $\min \|\mathbf{Ax} - \mathbf{b}\|^2$

8.4 Dynamic Programming

For sequential decision problems, **Bellman's Principle of Optimality**:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

The **Bellman equation** for discrete-time optimal control:

$$V^*(x_k) = \min_{u_k} [L(x_k, u_k) + V^*(f(x_k, u_k))] \quad (8.12)$$

where V^* is the optimal value function (cost-to-go).

Chapter 9

Discrete-Time Mathematics

Digital control systems operate on sampled data, requiring discrete-time analysis.

9.1 Sequences and Difference Equations

A **sequence** is a function defined on integers: $x[n]$ or x_n for $n \in \mathbb{Z}$.

First-order difference equation

$$x[n+1] = ax[n] + bu[n] \quad (9.1)$$

Solution:

$$x[n] = a^n x[0] + \sum_{k=0}^{n-1} a^{n-1-k} bu[k] \quad (9.2)$$

9.2 The Z-Transform

Definition 0.8: Z-Transform

The **Z-transform** of a sequence $x[n]$ is:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (9.3)$$

Common Z-Transform Pairs

Z-Transform Properties

9.3 Relationship Between s-Domain and z-Domain

For a sampled signal with sampling period T :

$$z = e^{sT} \quad (9.4)$$

Therefore:

$$s = \frac{1}{T} \ln z \quad (9.5)$$

Sequence	$x[n]$ for $n \geq 0$	$X(z)$
Unit impulse	$\delta[n]$	1
Unit step	$u[n] = 1$	$\frac{z}{z-1}$
Ramp	n	$\frac{z}{(z-1)^2}$
Exponential	a^n	$\frac{z}{z-a}$
Exponential ramp	na^n	$\frac{az}{(z-a)^2}$
Sine	$\sin(\omega n)$	$\frac{z \sin \omega}{z^2 - 2z \cos \omega + 1}$
Cosine	$\cos(\omega n)$	$\frac{z(z - \cos \omega)}{z^2 - 2z \cos \omega + 1}$

Table 9.1: Common Z-transform pairs.

Property	Sequence	Z-Domain
Linearity	$ax[n] + by[n]$	$aX(z) + bY(z)$
Time shift (delay)	$x[n - k]$	$z^{-k}X(z)$
Time advance	$x[n + 1]$	$zX(z) - zx[0]$
Convolution	$x[n] * y[n]$	$X(z)Y(z)$
Initial value	$x[0]$	$\lim_{z \rightarrow \infty} X(z)$
Final value	$\lim_{n \rightarrow \infty} x[n]$	$\lim_{z \rightarrow 1} (z - 1)X(z)$

Table 9.2: Z-transform properties.

Stability in z-Domain

A discrete-time system is stable if and only if all poles of its transfer function $G(z)$ lie **inside the unit circle** $|z| < 1$.

The imaginary axis in the s -plane maps to the unit circle in the z -plane:

- Left half-plane ($\text{Re}(s) < 0$) $\rightarrow$ inside unit circle ($|z| < 1$)
- Right half-plane ($\text{Re}(s) > 0$) $\rightarrow$ outside unit circle ($|z| > 1$)
- $s = 0 \rightarrow z = 1$
- $s = \pm j\omega_s/2 \rightarrow z = -1$ (Nyquist frequency)

9.4 Discrete-Time State-Space

The discrete-time state-space representation:

$$\mathbf{x}[k + 1] = \mathbf{A}_d \mathbf{x}[k] + \mathbf{B}_d \mathbf{u}[k] \quad (9.6)$$

$$\mathbf{y}[k] = \mathbf{C}_d \mathbf{x}[k] + \mathbf{D}_d \mathbf{u}[k] \quad (9.7)$$

Conversion from continuous-time (exact discretization with zero-order hold):

$$\mathbf{A}_d = e^{\mathbf{A}T} \quad (9.8)$$

$$\mathbf{B}_d = \int_0^T e^{\mathbf{A}\tau} d\tau \cdot \mathbf{B} = \mathbf{A}^{-1}(e^{\mathbf{A}T} - \mathbf{I})\mathbf{B} \quad (9.9)$$

Chapter 10

Special Functions and Signals

10.1 Unit Step Function

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \quad (10.1)$$

The discrete-time unit step:

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases} \quad (10.2)$$

10.2 Impulse (Dirac Delta) Function

The **Dirac delta function** is defined by:

$$\delta(t) = 0 \quad \text{for } t \neq 0 \quad (10.3)$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (10.4)$$

Sifting property

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0) \quad (10.5)$$

Relationship to step function:

$$\delta(t) = \frac{du(t)}{dt} \quad (10.6)$$

10.3 Ramp Function

$$r(t) = t \cdot u(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases} \quad (10.7)$$

10.4 Sinusoidal Functions

General sinusoid:

$$x(t) = A \sin(\omega t + \phi) = A \cos(\omega t + \phi - \pi/2) \quad (10.8)$$

where:

- A = amplitude
- $\omega = 2\pi f$ = angular frequency (rad/s)
- f = frequency (Hz)
- ϕ = phase (rad)
- $T = 1/f = 2\pi/\omega$ = period (s)

Complex exponential representation

$$A \cos(\omega t + \phi) = \operatorname{Re}\{Ae^{j(\omega t + \phi)}\} \quad (10.9)$$

10.5 Exponential Functions

Real exponential

$$x(t) = Ae^{\sigma t} \quad (10.10)$$

- $\sigma < 0$: decaying
- $\sigma > 0$: growing
- $\sigma = 0$: constant

Complex exponential (damped sinusoid)

$$x(t) = Ae^{\sigma t} \cos(\omega t + \phi) \quad (10.11)$$

Chapter 11

Numerical Methods

11.1 Numerical Integration of ODEs

Euler's Method (Forward)

$$x_{n+1} = x_n + hf(t_n, x_n) \quad (11.1)$$

Runge-Kutta 4th Order (RK4)

$$k_1 = f(t_n, x_n) \quad (11.2)$$

$$k_2 = f(t_n + h/2, x_n + hk_1/2) \quad (11.3)$$

$$k_3 = f(t_n + h/2, x_n + hk_2/2) \quad (11.4)$$

$$k_4 = f(t_n + h, x_n + hk_3) \quad (11.5)$$

$$x_{n+1} = x_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (11.6)$$

11.2 Root Finding

Newton-Raphson

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (11.7)$$

Bisection Method

For continuous f with $f(a)f(b) < 0$, repeatedly halve the interval containing the root.

11.3 Numerical Linear Algebra

Key algorithms:

- **Gaussian elimination:** $O(n^3)$ for solving $\mathbf{Ax} = \mathbf{b}$
- **QR iteration:** Computing eigenvalues
- **SVD algorithms:** Golub-Kahan bidiagonalization

- **Iterative methods:** Conjugate gradient, GMRES for large sparse systems

Chapter 12

Exercises

Exercise 12.1. Convert the following complex numbers to polar form:

(a) $z_1 = -3 + 4j$

(b) $z_2 = 2 - 2j$

(c) $z_3 = -5$

Exercise 12.2. Find the Laplace transform of:

(a) $f(t) = 3e^{-2t} \sin(4t)$

(b) $f(t) = t^2 e^{-t}$

(c) $f(t) = (1 - e^{-t})u(t)$

Exercise 12.3. Find the inverse Laplace transform of:

(a) $F(s) = \frac{5}{s(s+3)}$

(b) $F(s) = \frac{s+1}{s^2+4s+13}$

(c) $F(s) = \frac{1}{(s+1)^2(s+2)}$

Exercise 12.4. For the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$:

(a) Find the eigenvalues

(b) Find the eigenvectors

(c) Compute $e^{\mathbf{A}t}$

(d) Is the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ stable?

Exercise 12.5. Linearize the nonlinear system $\dot{x} = x^2 - u$ about the equilibrium point where $x_0 = 1$, $u_0 = 1$.

Exercise 12.6. Find the Z-transform and region of convergence for:

(a) $x[n] = (0.5)^n u[n]$

(b) $x[n] = n(0.8)^n u[n]$

(c) $x[n] = \cos(0.2\pi n)u[n]$

Exercise 12.7. Using gradient descent, find the minimum of $f(x, y) = x^2 + 2y^2 - 2xy - 2x$ starting from $(0, 0)$ with step size $\alpha = 0.1$. Perform 5 iterations.

Exercise 12.8. For a random variable $X \sim \mathcal{N}(5, 4)$:

(a) Find $P(X > 7)$

(b) Find $P(3 < X < 7)$

(c) Find the value x such that $P(X < x) = 0.95$